Indian Institute of Technology Roorkee | Department of Computer Science and Engineering

HVAC Service Planner

Using machine learning for HVAC system providers such as Aire Serv

Demand forecasting, failure prediction and customer retention

CSC-505 Machine Learning | Group 1

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Under the guidance of

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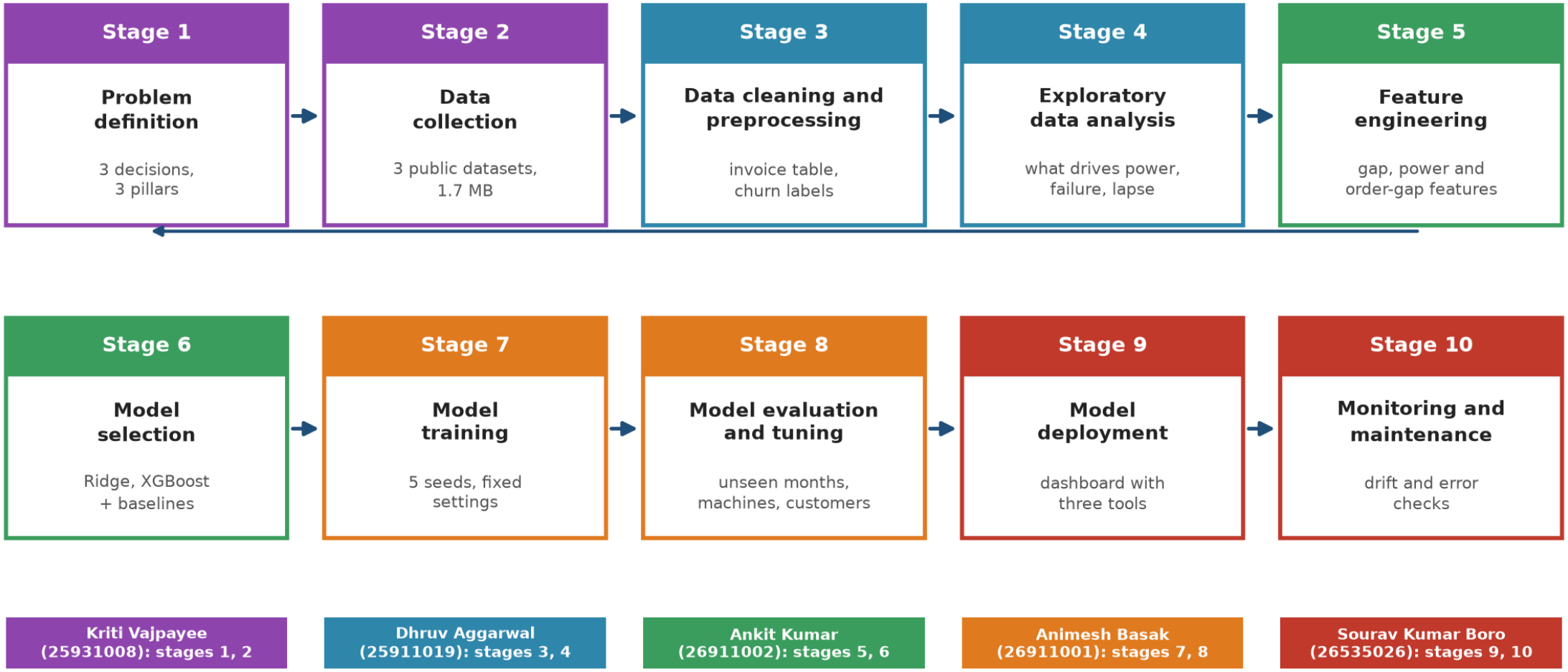
Professor, Department of Computer Science and Engineering, IIT Roorkee

October 2026

Ten lifecycle stages, five authors, one product

Summary
Whole group

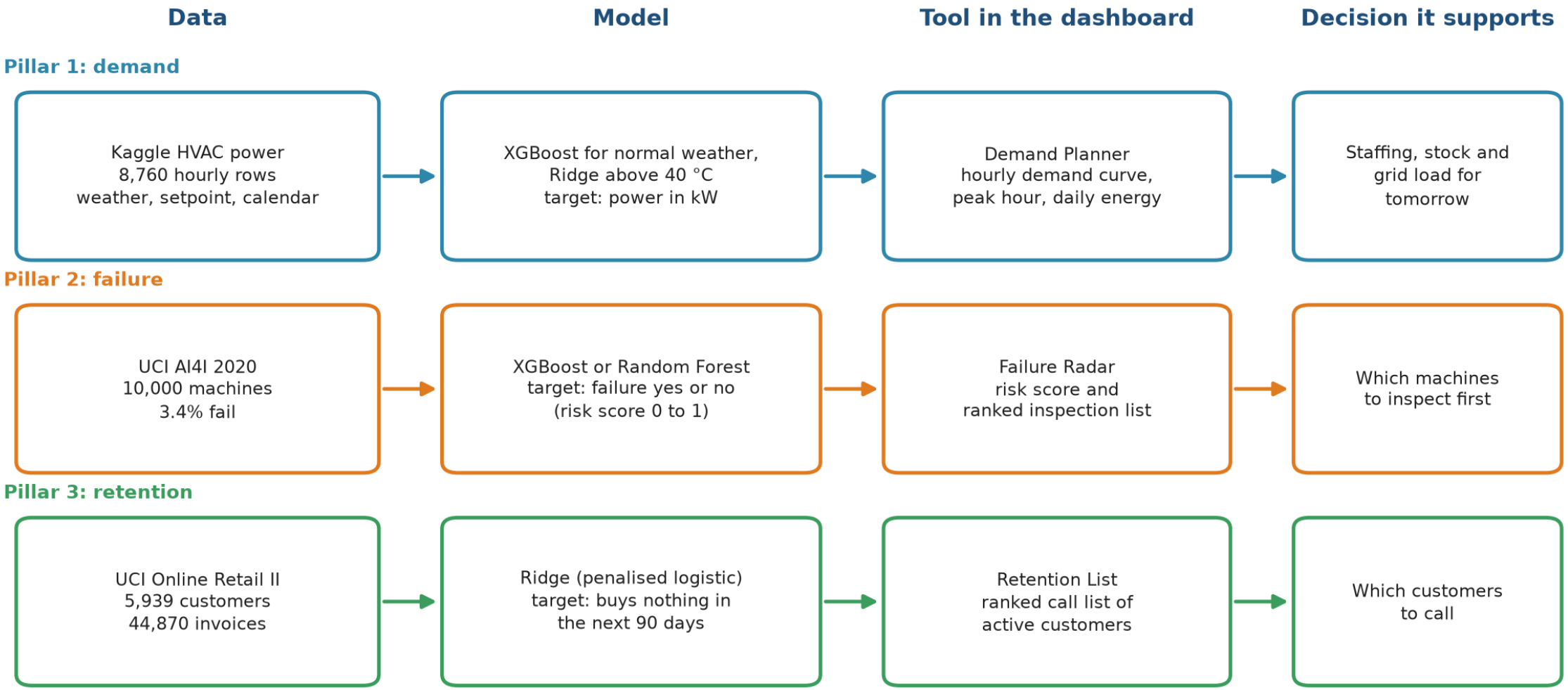
The ten lifecycle stages and who wrote each one



One dashboard, three decisions

Stage 1: Problem definition
Kriti Vajpayee

HVAC Service Planner: from data to a decision



Each question becomes a precise learning task

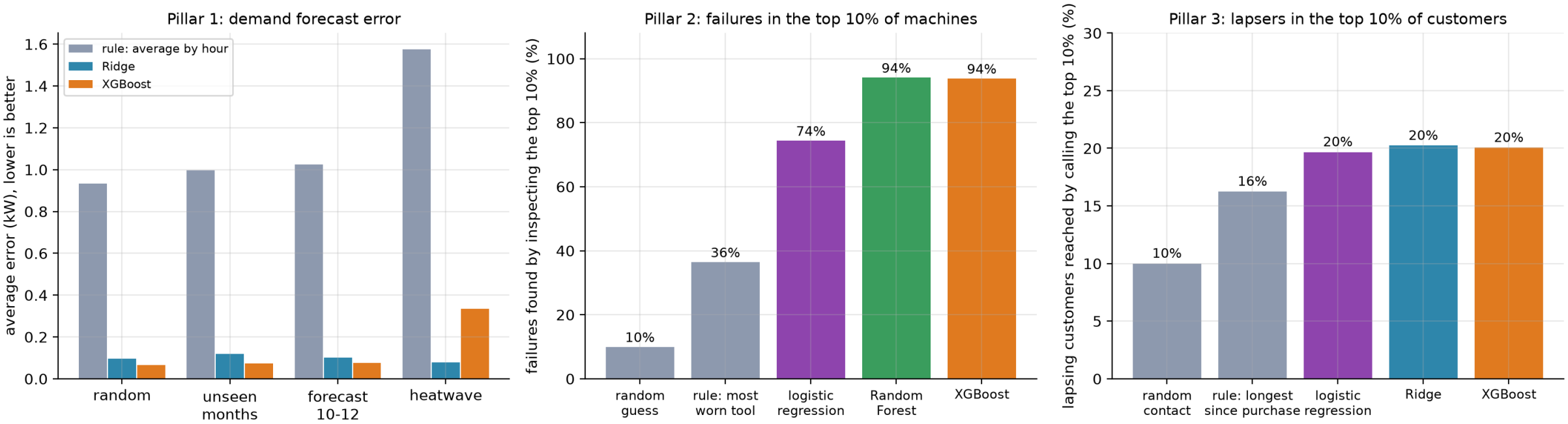
Stage 1: Problem definition
Kriti Vajpayee

	Pillar 1: demand	Pillar 2: failure	Pillar 3: retention
Task type	Regression (a number)	Binary classification, rare event	Binary classification
Unit of prediction	One building-hour	One machine	One active customer at a cutoff date
Inputs	Outdoor and indoor temperature, humidity, hour, month, mode	Air and process temperature, speed, torque, tool wear, type	Purchase history before the cutoff
Output	Power in kW	Failure yes or no, risk score	Lapses yes or no, risk score
Looking ahead	Same hour, weather forecast as input	The recorded failure label	90 days after the cutoff
Main metric	Mean absolute error in kW	PR-AUC; failures found in the top 10%	ROC-AUC; lapsers found in the top 10%
Why this metric	Average miss, easy to read	Failures are 3.4% of machines, so accuracy is meaningless	A small call budget means the top of the list matters

The bar to beat: what a rule of thumb achieves

Stage 1: Problem definition
Kriti Vajpayee

The bar to beat: what a rule of thumb achieves, against our models (unseen data)



- For each pillar we picked the simplest sensible rule a company could use today.
- A model must beat it, and also beat a statistical and a classic ML baseline.

Three public datasets, 1.7 MB in total

Stage 2: Data collection
Kriti Vajpayee

The three datasets

Pillar 1: HVAC power (Kaggle)

felixpradoh/hvac-power-consumption
hvac_power_consumption.csv

size: 8,760 rows, 8 columns (0.5 MB)
period: one year, hourly (2025)

columns: month, hour, outdoor temp, indoor temp, humidity, HVAC mode, power in kW

target: power_kw
range 0.10 to 4.01 kW,
mean 1.92 kW

caveat: very regular, probably simulated

Pillar 2: machine failures (UCI 601)

AI4I 2020 Predictive Maintenance
ai4i2020.csv

size: 10,000 machines, 14 columns (0.5 MB)
sensor readings and a failure label

columns: type L/M/H, air temp, process temp, speed, torque, tool wear

target: machine failure (yes or no)
339 failures = 3.39%

caveat: synthetic data about machine tools, not HVAC units

Pillar 3: customers (UCI 502)

Online Retail II
retail_invoices.csv.gz

size: 44,870 invoices, 5,939 customers
(0.8 MB; the original is 1.07 million lines, 45 MB)
period: Dec 2009 to Dec 2011

columns: customer, invoice, date, value, items, products, country, is_return

target: built by us (no purchase in the next 90 days)

caveat: a UK online gift shop, not a service company

All three are public. Together they are 1.7 MB in hvac_ml/data/.

The datasets at a glance, with sources

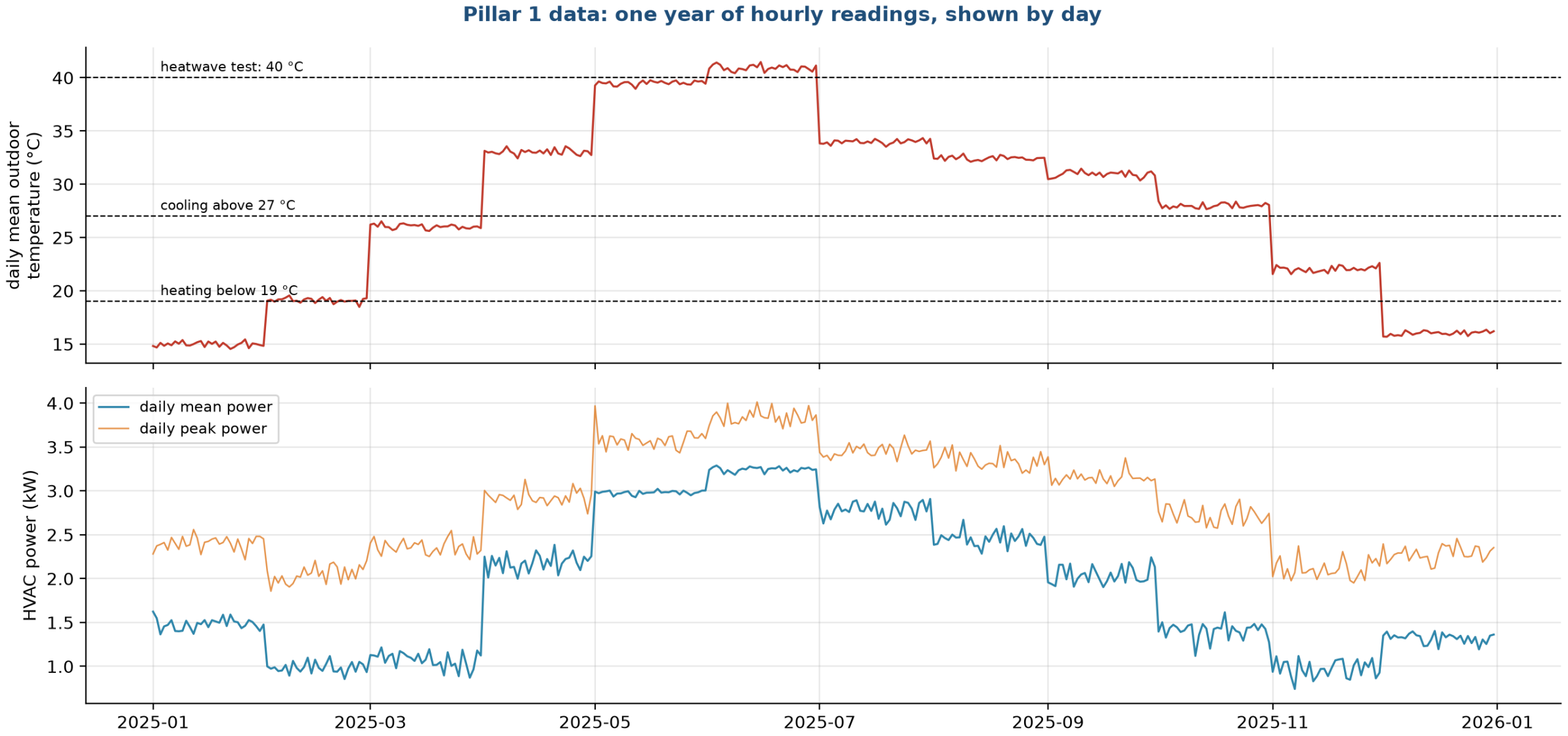
Stage 2: Data collection
Kriti Vajpayee

Pillar	Dataset	Rows	Size	Target	Source
1. Demand	HVAC power consumption, hourly 2025	8,760	0.49 MB	Power in kW	kaggle.com/datasets/felixpradoh/hvac-power-consumption
2. Failure	AI4I 2020 Predictive Maintenance	10,000	0.52 MB	Machine failure, 3.39%	archive.ics.uci.edu/dataset/601
3. Churn	Online Retail II (one row per invoice)	44,870 invoices	0.76 MB	No purchase in the next 90 days (built by us)	archive.ics.uci.edu/dataset/502

Caveats: the HVAC file looks simulated, AI4I is synthetic data about machine tools, and Online Retail II is a UK online gift shop.

A year of hourly readings that move in monthly steps

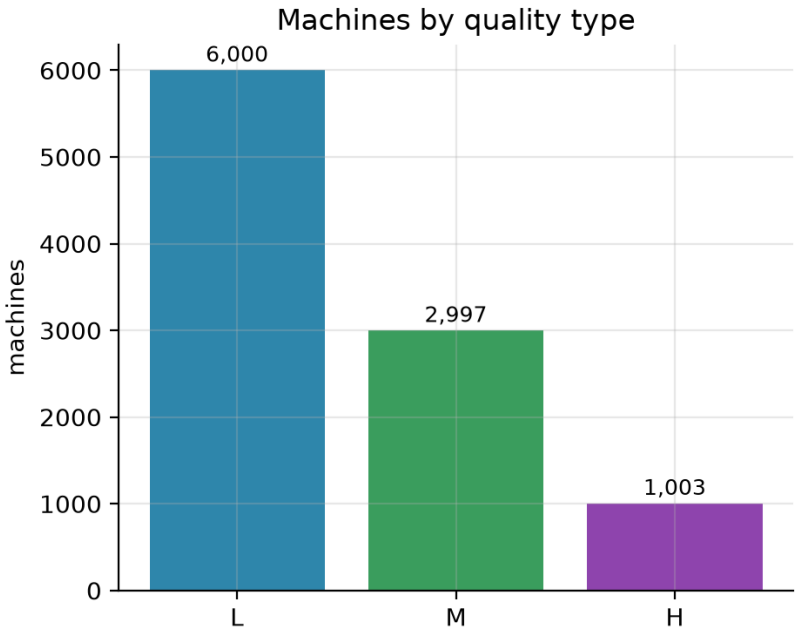
Stage 2: Data collection
Kriti Vajpayee

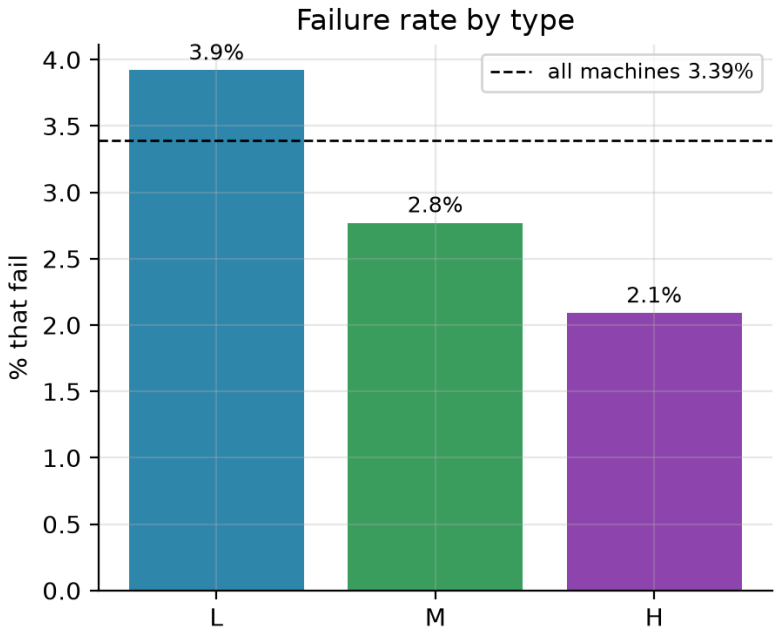


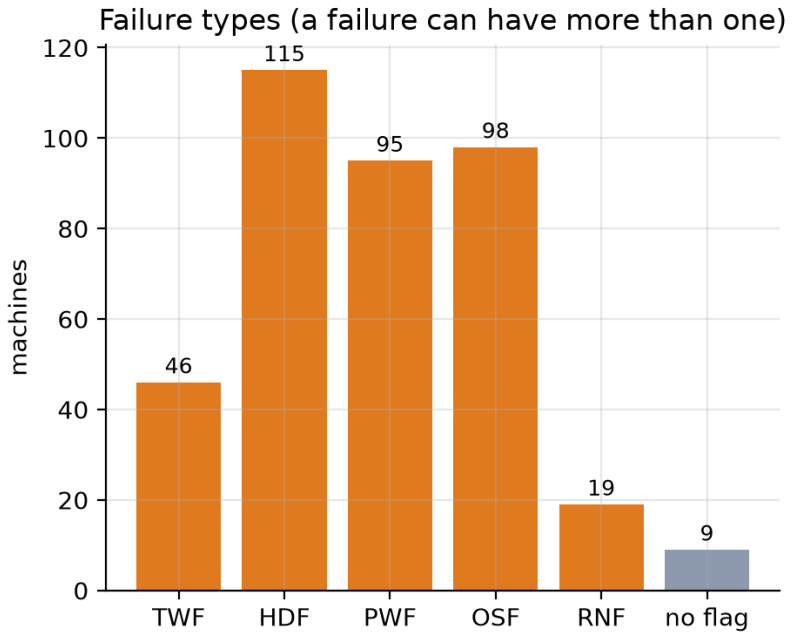
AI4I: 339 failures in 10,000 machines

Stage 2: Data collection
Kriti Vajpayee

Pillar 2 data: 10,000 machines, 339 failures





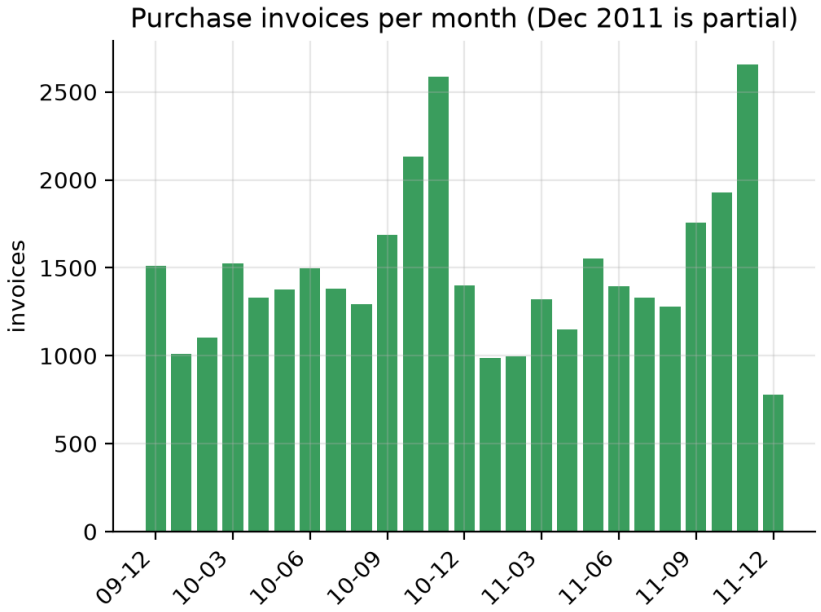


- Low-quality machines fail most.
- Heat dissipation, overstrain and power failures dominate.
- The failure-type columns are never used as inputs.

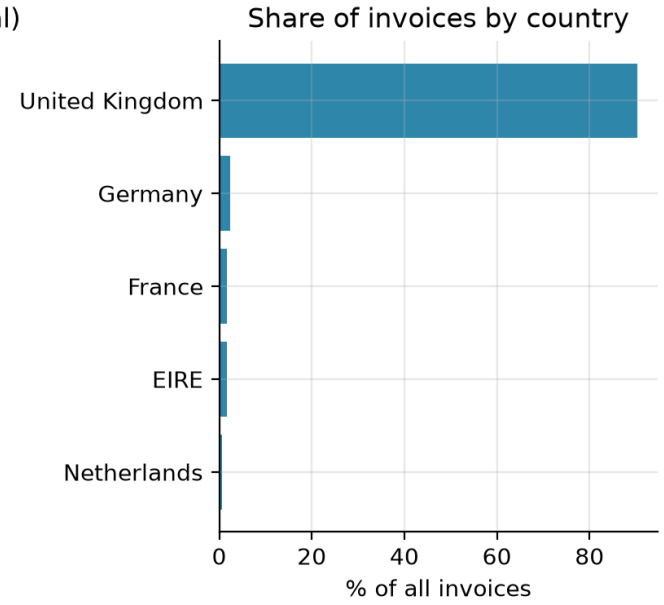
Online Retail II: seasonal peaks, mostly UK customers

Stage 2: Data collection
Kriti Vajpayee

Pillar 3 data: 44,870 invoices from 5,939 customers; the UK is 90% of invoices; 7,901 are returns



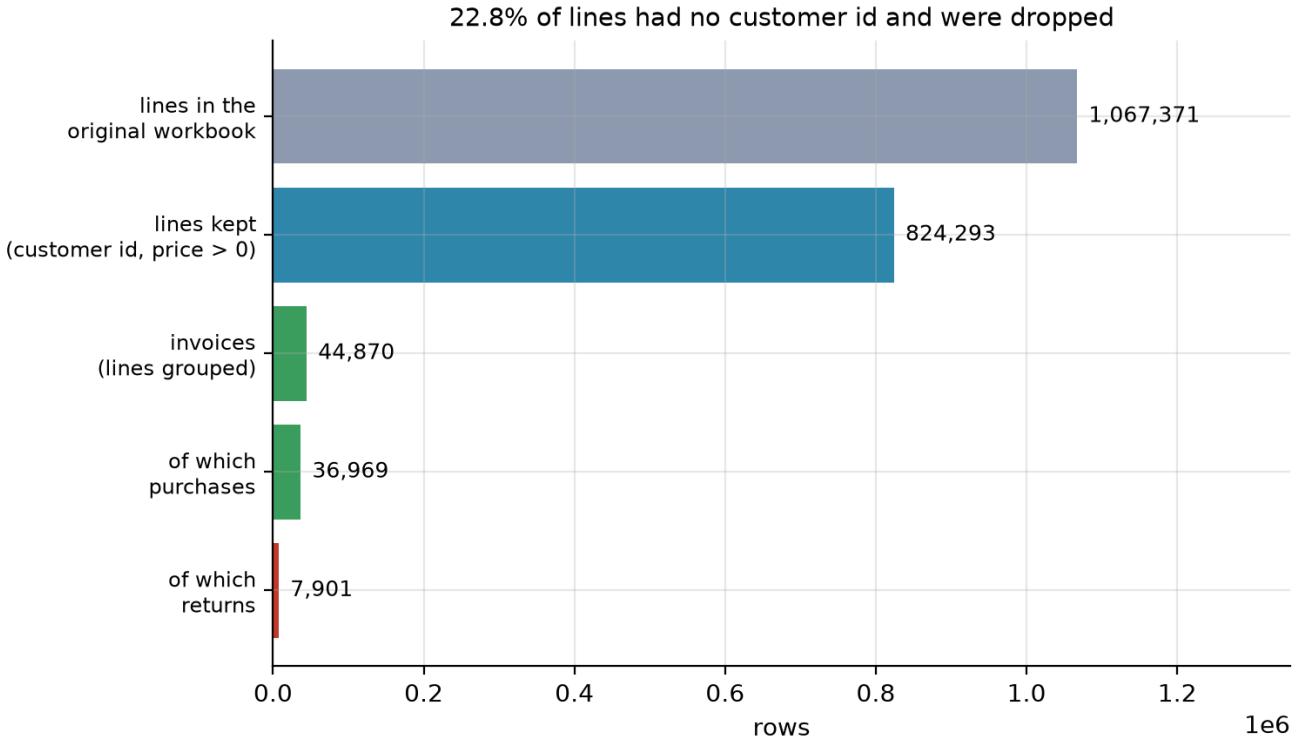


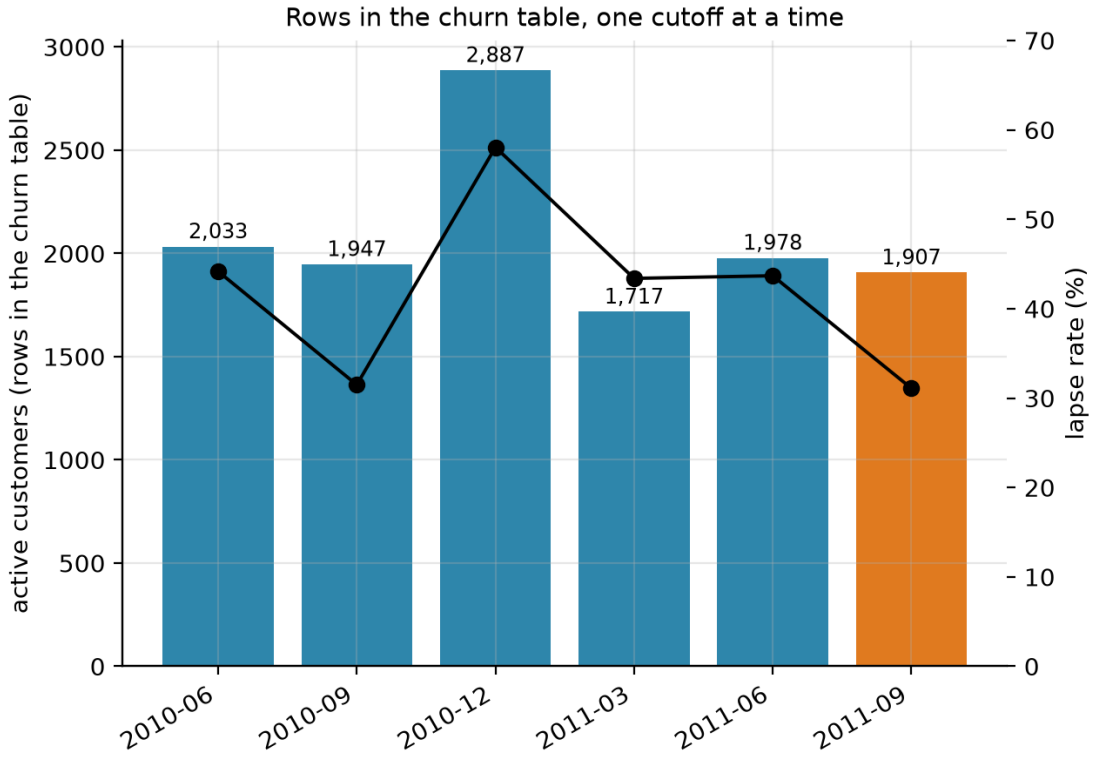


- Invoices and active customers peak in October and November.
- December 2011 is partial.
- This seasonality affects how we read churn.

From 1.07 million lines to 44,870 invoices

Stage 3: Data cleaning
Dhruv Aggarwal

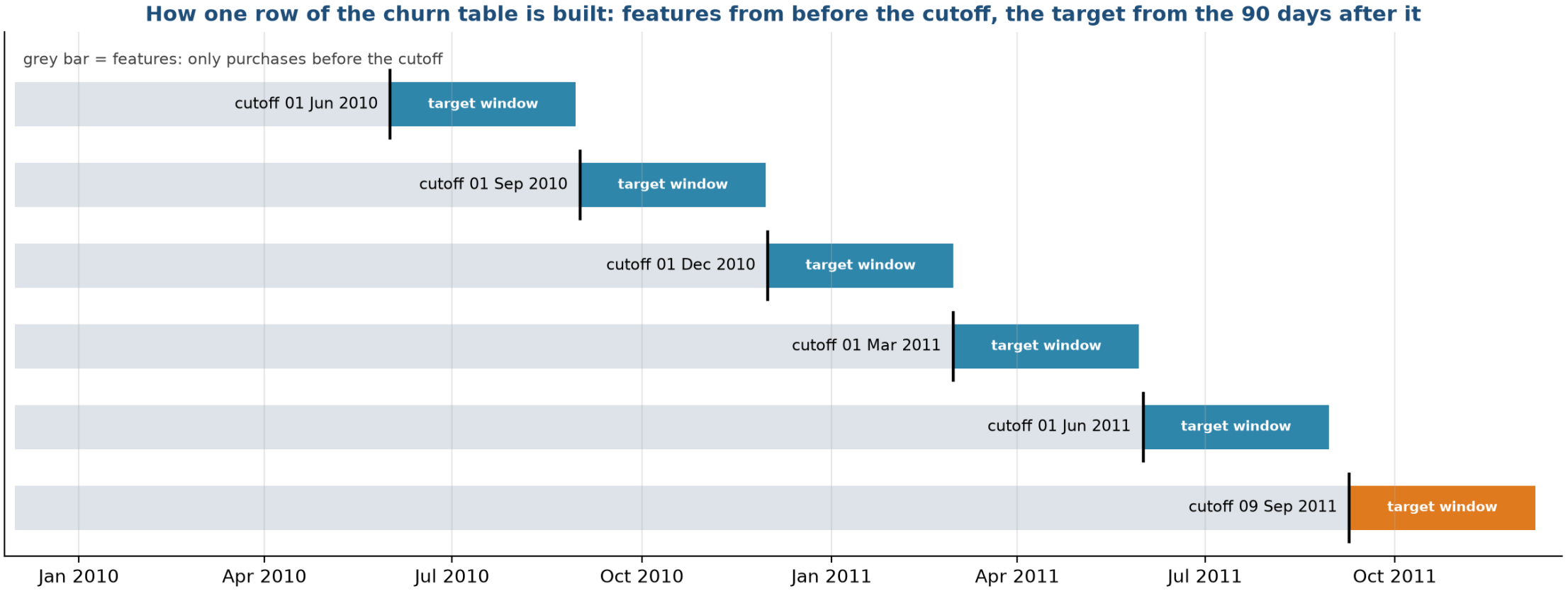




- We drop lines without a customer id (22.8%) and lines with a price of zero or less, then group by invoice.
- The right panel shows rows per churn snapshot and the lapse rate.

The churn target is built from the future

Stage 3: Data cleaning
Dhruv Aggarwal



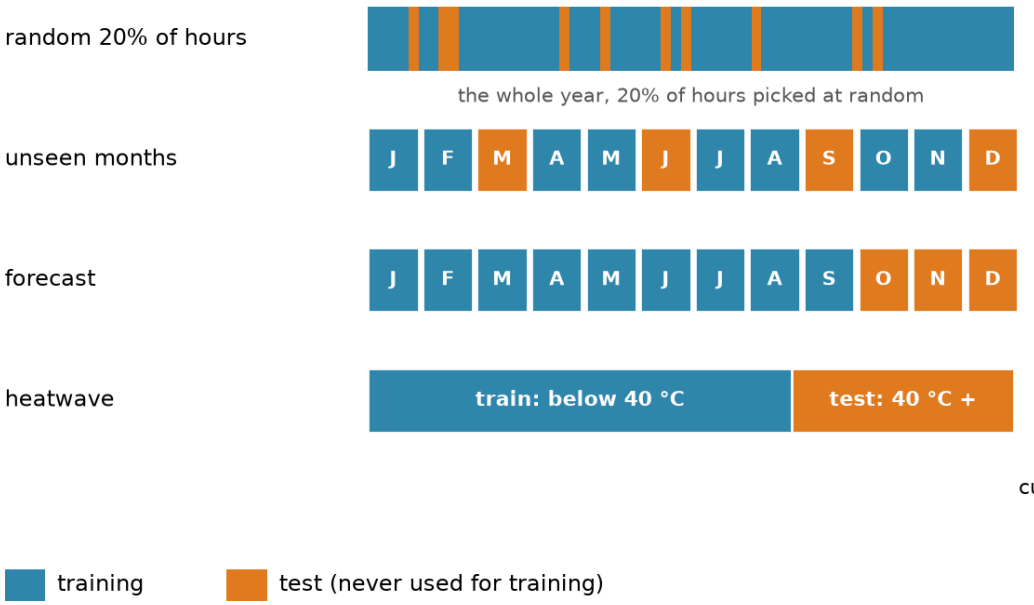
Blue rows (training): the five earlier cutoffs, training customers only (80%). Orange row (test): one later cutoff, unseen customers only (20%). The last training window ends 30 Aug 2011, before the test cutoff.

A customer is included only if they bought in the 90 days before the cutoff. Target = 1 when they buy nothing in the 90 days after it.

Every test hides data the model never saw

How each pillar is split into training and test data

Pillar 1: demand (four tests)



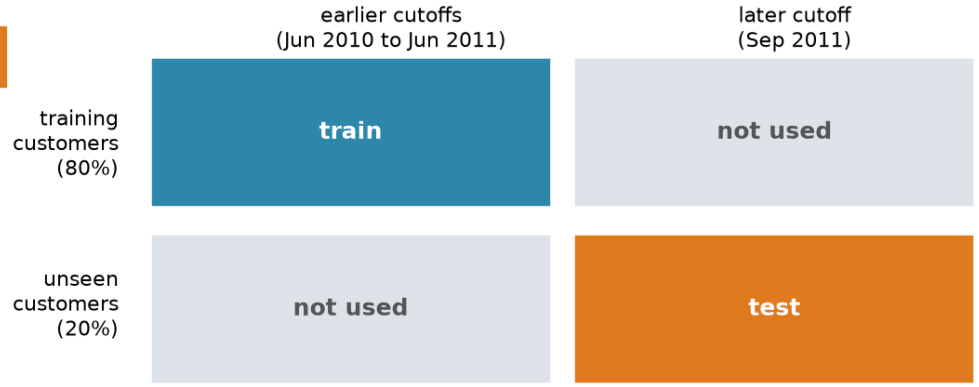
Reference models: the training mean and the average by hour of day, scored on the same test rows.

Pillar 2: failure



stratified: both parts keep the same 3.4% failure share; repeated with 5 seeds (a different split each time)

Pillar 3: churn (time and customers)



No customer appears in both training and test, and every training window ends before the test cutoff.

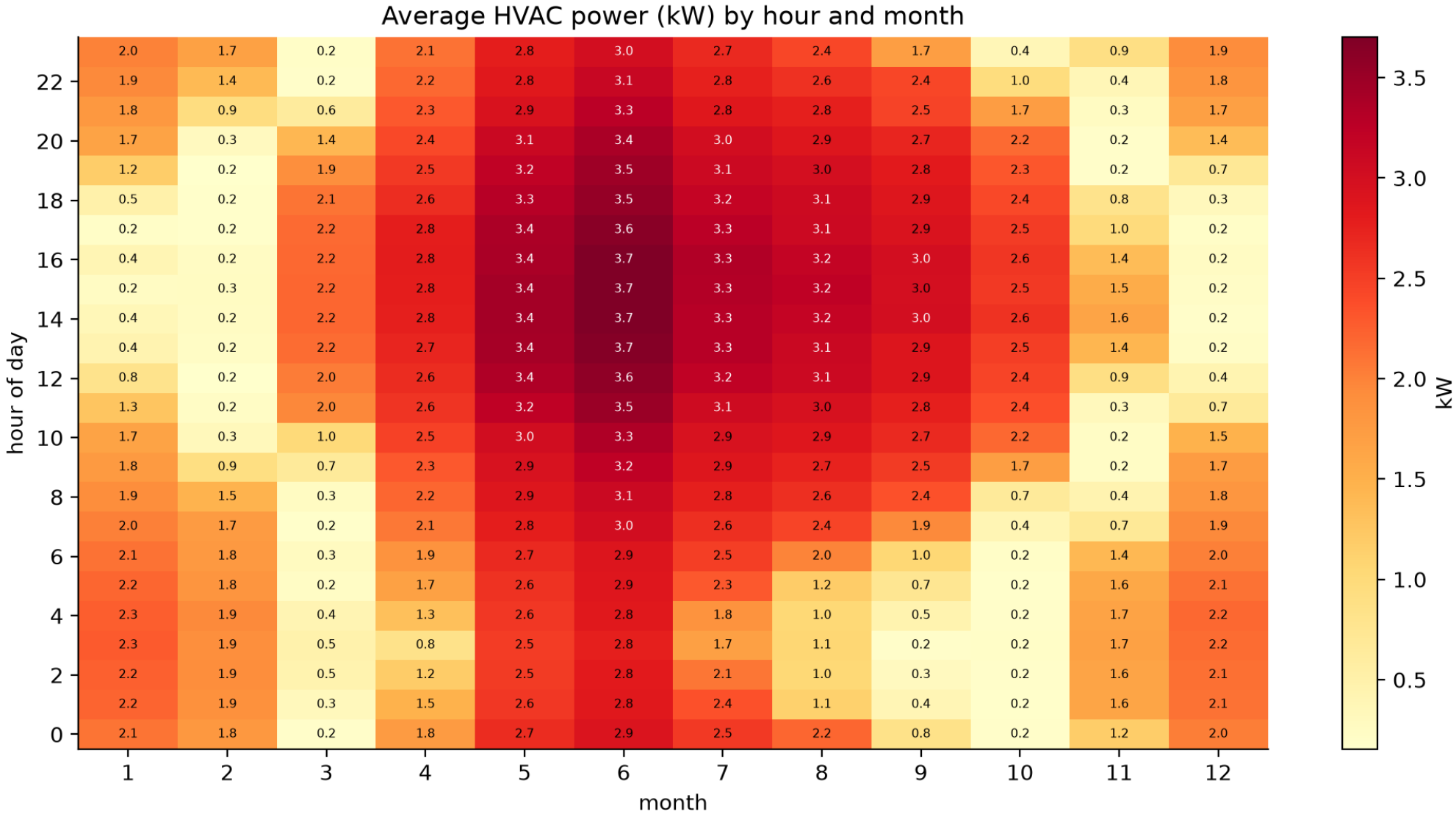
Seven guards against leakage

Stage 3: Data cleaning
Dhruv Aggarwal

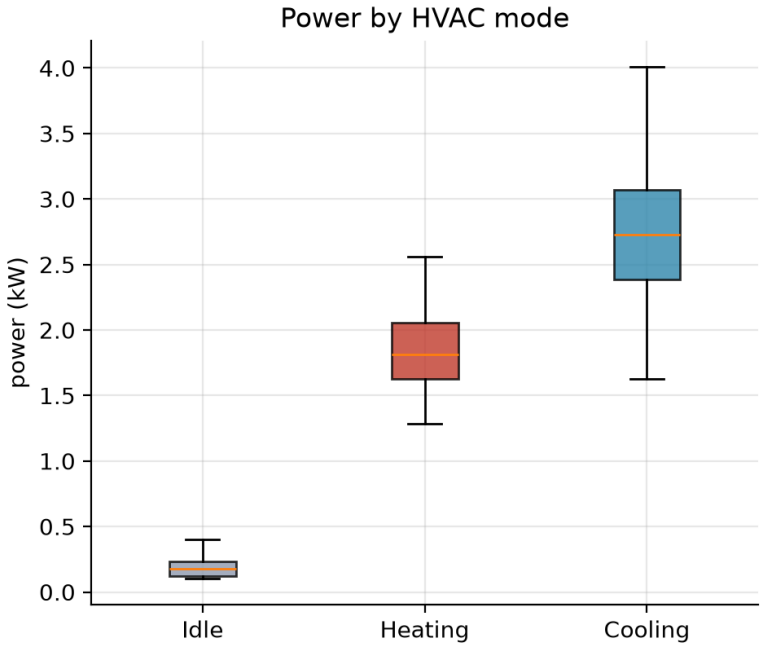
Risk	What we did	Check
Features that use the future (churn)	Features look only at invoices before the cutoff	Automated test deletes later invoices
Target built from the past (churn)	Target looks only at invoices after the cutoff	Automated test
Same customer in training and test	Customers are split before rows are built	Automated test
Overlapping windows	Last training window ends before the test cutoff	Automated test
Answer hidden in the inputs (failure)	The five failure-type columns are never inputs	Automated test
Test statistics in the pipeline	Imputing and scaling sit inside the model pipeline	By design
Threshold chosen on test data	Threshold comes from out-of-fold training scores	By design

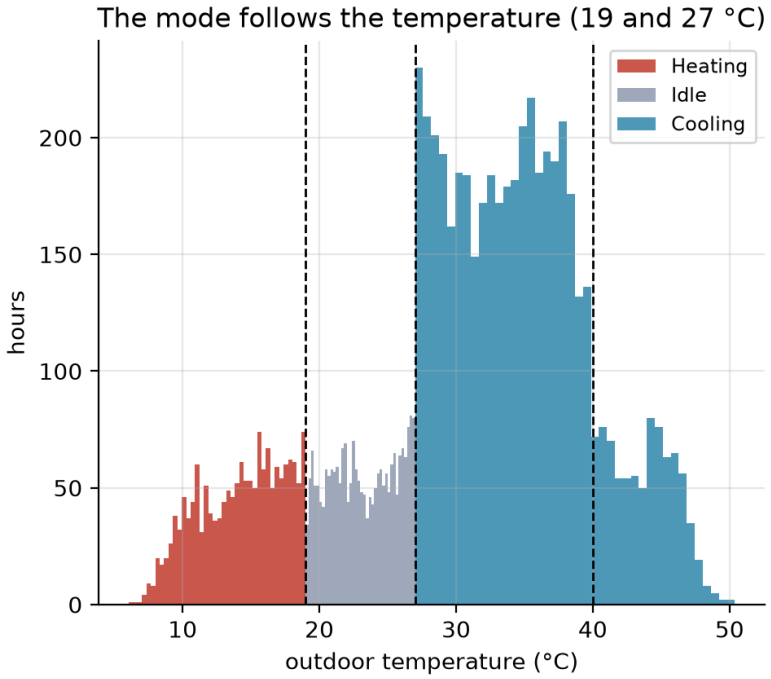
Power depends on the month first, the hour second

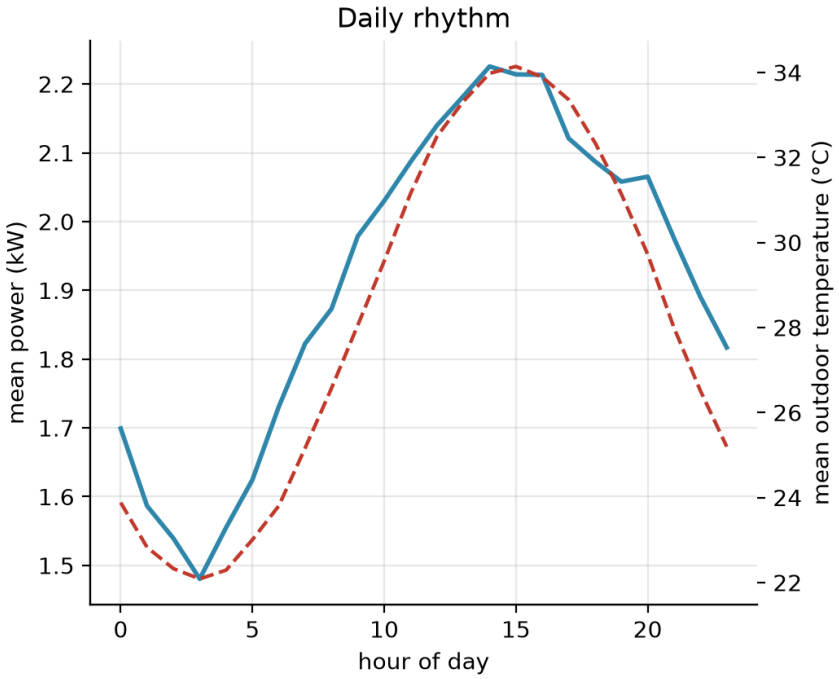
Stage 4: Data exploration
Dhruv Aggarwal



Mode follows temperature; power follows the gap

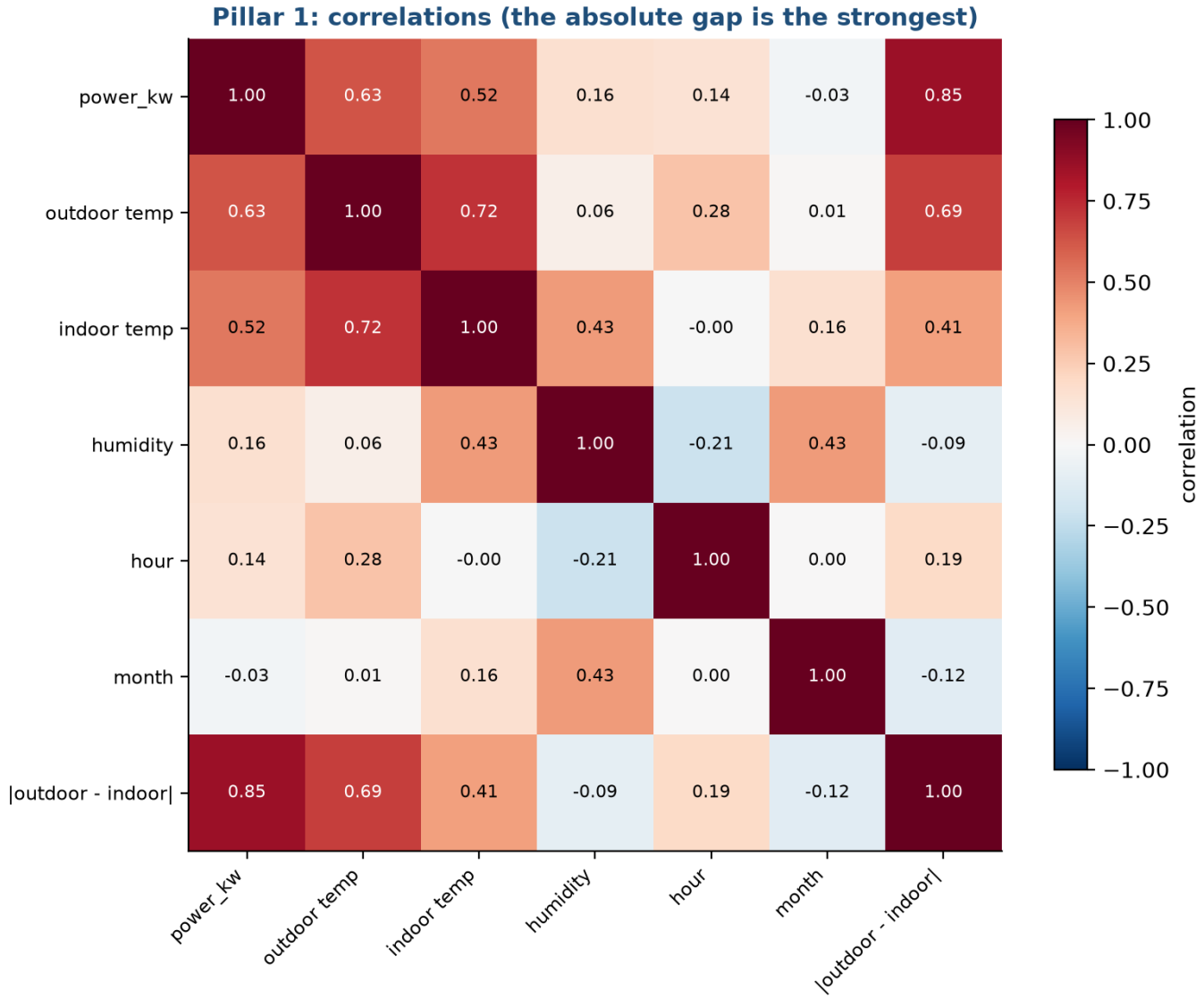






- Idle 0.19 kW, heating 1.84 kW, cooling 2.73 kW.
- The 824 hours at 40 degrees or more are all in cooling mode.

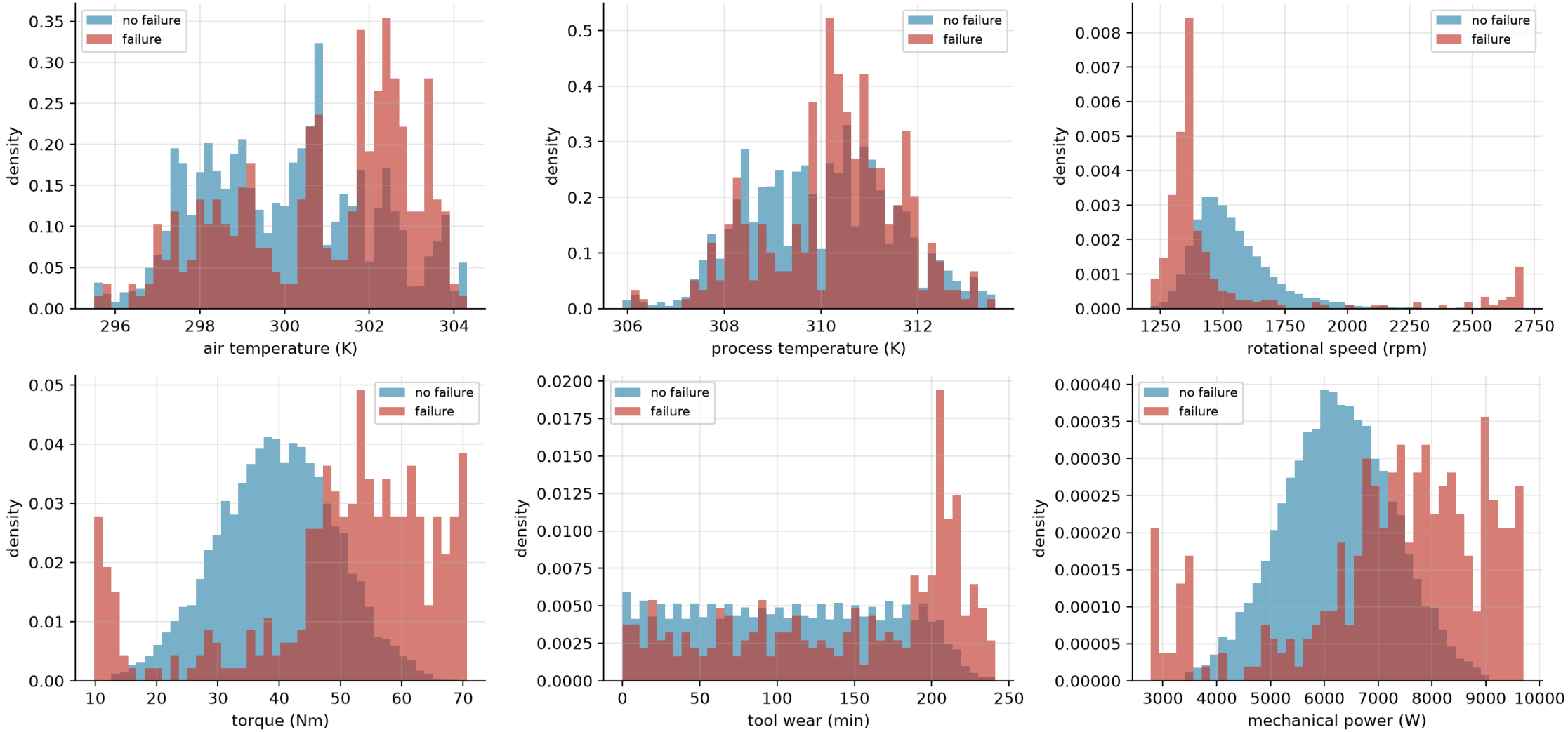
The absolute gap explains power better than temperature



Failing machines sit at both ends of torque and power

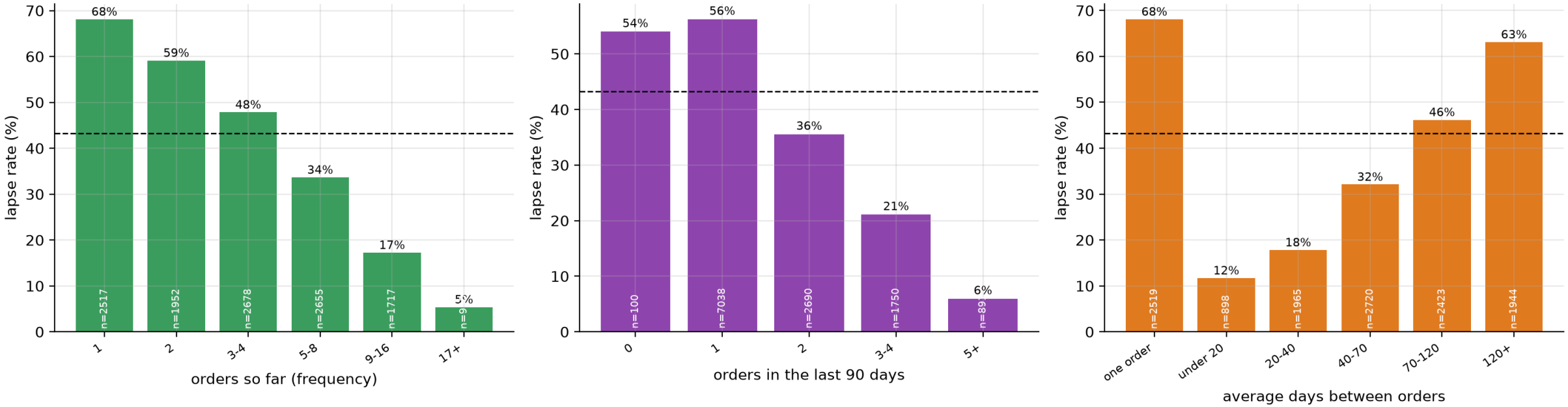
Stage 4: Data exploration
Dhruv Aggarwal

Pillar 2: each sensor reading, machines that failed (red) against those that did not (blue)



Customers who order often and regularly rarely lapse

Pillar 3: who lapses (all six cutoffs combined; dashed line = overall lapse rate)



- Lapse falls from 68% with one order to 5% with 17 or more.
- A short order rhythm predicts staying.
- Recency, the basis of the usual rule, is the weakest signal.

What the exploration told us to do

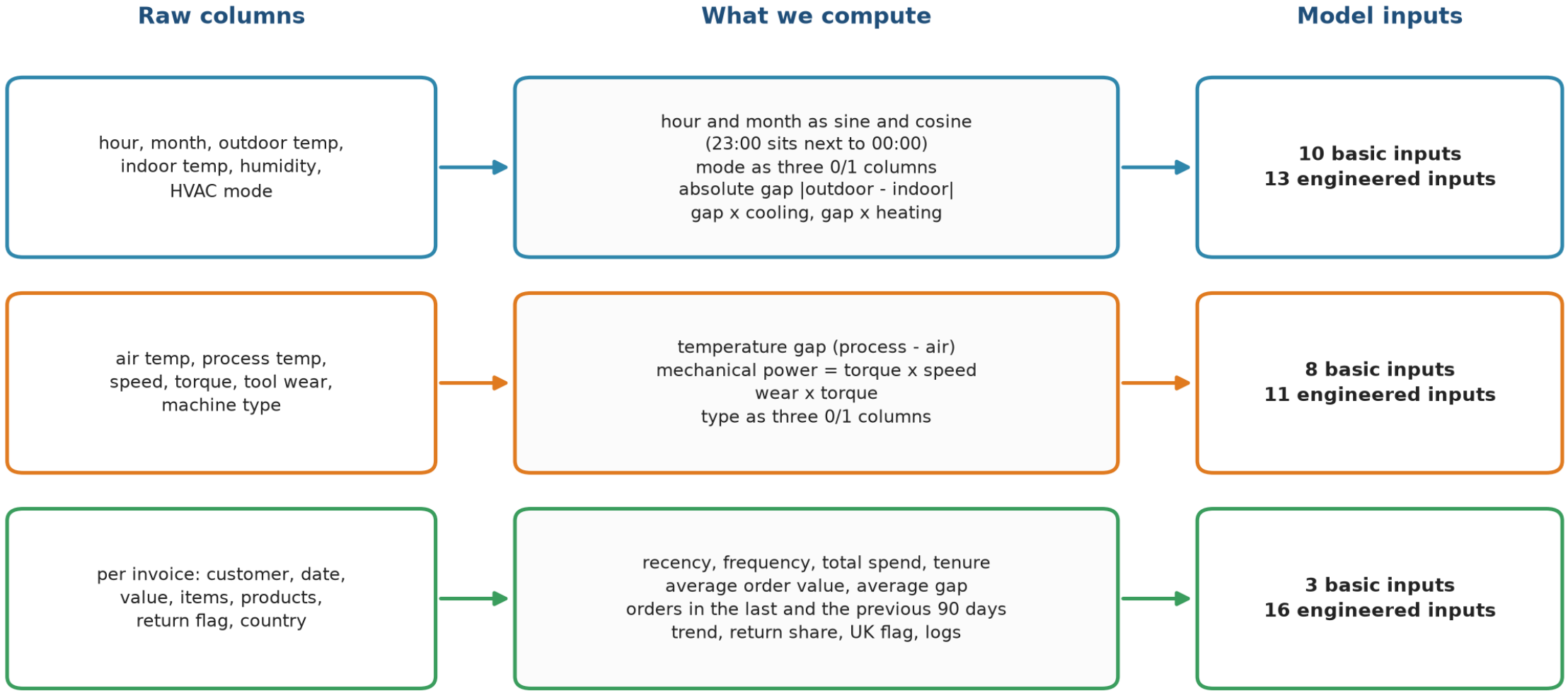
Stage 4: Data exploration
Dhruv Aggarwal

Pillar	Finding	Number	Consequence
1	Power follows the gap between outdoor and indoor temperature	correlation 0.85	Add the absolute gap; a linear model on it can extrapolate
1	Idle hours draw 0.19 kW, cooling 2.73 kW	by mode	Keep mode as an input
1	824 hours reach 40 °C or more, all in cooling mode	9.4% of hours	Use them as the heatwave test
2	Failure is U-shaped in torque, speed and power	17% to 18% in top decile, against 3.4%	Trees should beat linear models
2	No single input correlates strongly with failure	at most 0.19	Combine inputs; use PR-AUC
3	Lapse falls from 68% to 5% as order count rises	frequency	The signal is in the order history
3	Recency is the weakest of the main signals	correlation 0.16	The recency rule should be a weak baseline
3	Money columns are very skewed	top 1% hold 32% of spend	Use log versions

Engineered inputs come from the physics and the business

Stage 5: Feature engineering
Ankit Kumar

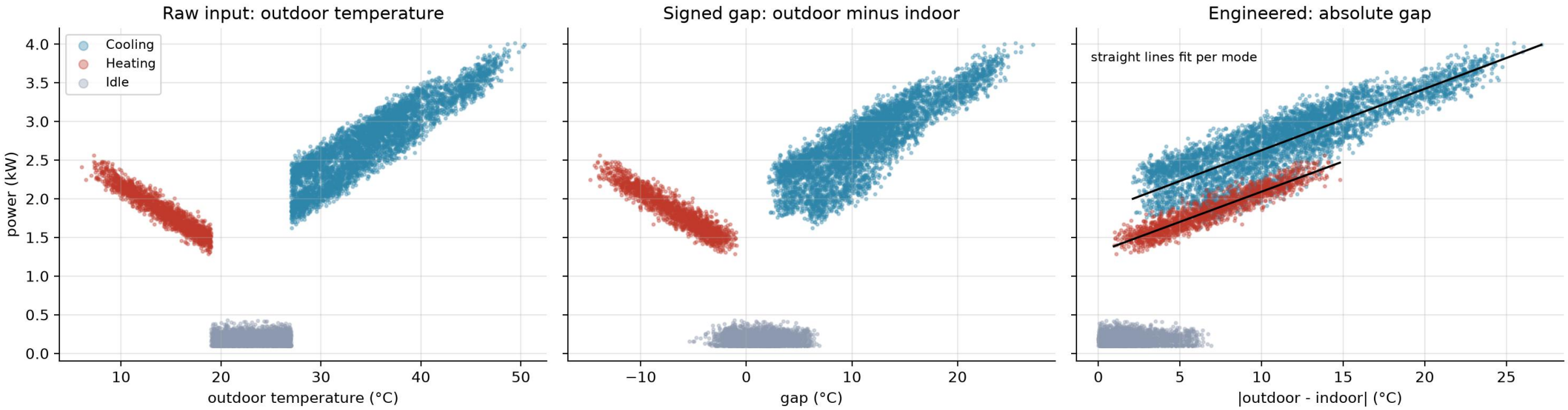
From raw columns to model inputs



Folding the V: absolute gap turns two slopes into one

Stage 5: Feature engineering
Ankit Kumar

Why the absolute gap helps: power is a V-shape in temperature but nearly a straight line in the gap

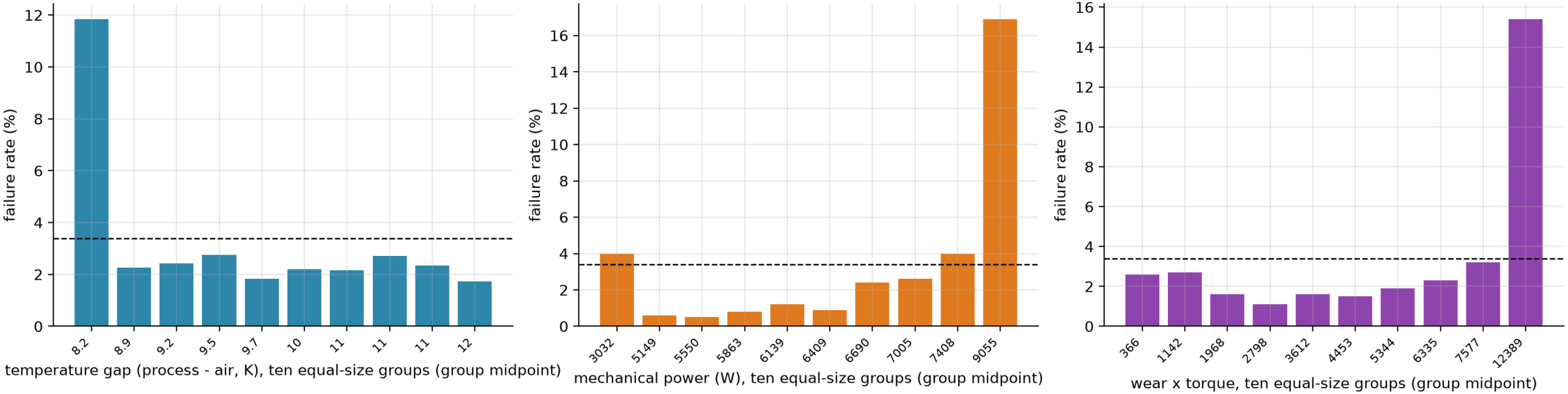


- Power against temperature is a V.
- Against the absolute gap it becomes a single rising band, which a straight line follows well.

Engineered failure inputs expose the thresholds

Stage 5: Feature engineering
Ankit Kumar

Pillar 2: the engineered inputs separate failures better than the raw sensors (dashed = 3.4% average)

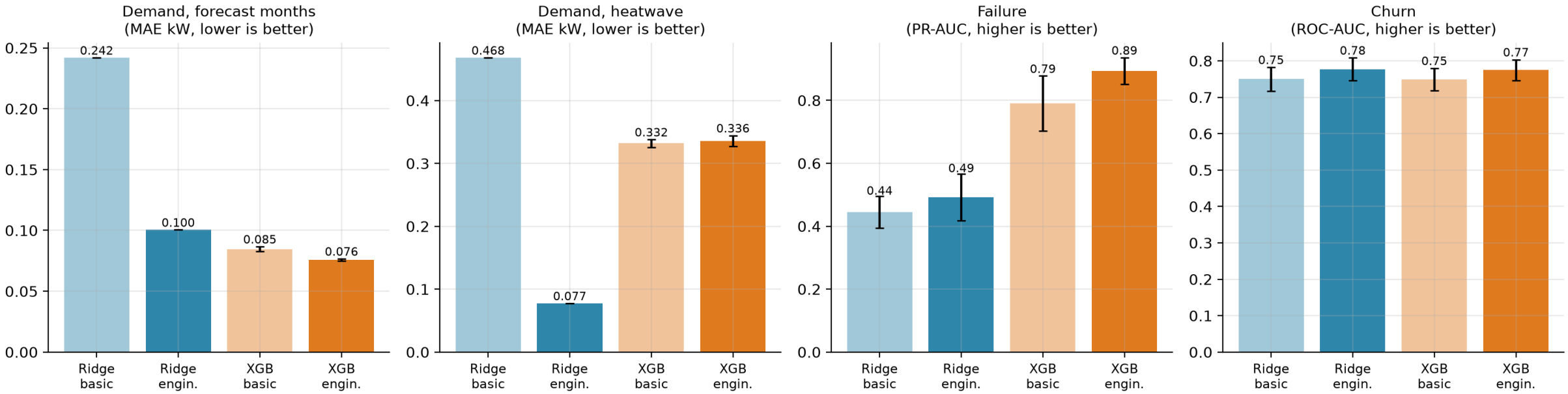


- The lowest temperature-gap tenth fails 12% of the time, the top mechanical-power tenth 17% and the top wear-times-torque tenth 15%.

Engineering halves Ridge error, and does little for trees

Stage 5: Feature engineering
Ankit Kumar

What the engineered features add (light bars: basic inputs, dark bars: engineered inputs)



- Linear models need the gap handed to them.
- XGBoost builds such shapes itself by splitting.

Basic against engineered inputs, on unseen data

Stage 5: Feature engineering
Ankit Kumar

Test	Ridge, basic	Ridge, engineered	XGBoost, basic	XGBoost, engineered
Demand, random split (MAE, kW)	0.182	0.096	0.065	0.065
Demand, unseen months (MAE, kW)	0.224	0.118	0.070	0.072
Demand, forecast (MAE, kW)	0.242	0.100	0.085	0.076
Demand, heatwave (MAE, kW)	0.468	0.077	0.332	0.336
Failure (PR-AUC)	0.445	0.492	0.790	0.893
Churn (ROC-AUC)	0.750	0.777	0.749	0.775
Churn (PR-AUC)	0.547	0.578	0.529	0.580

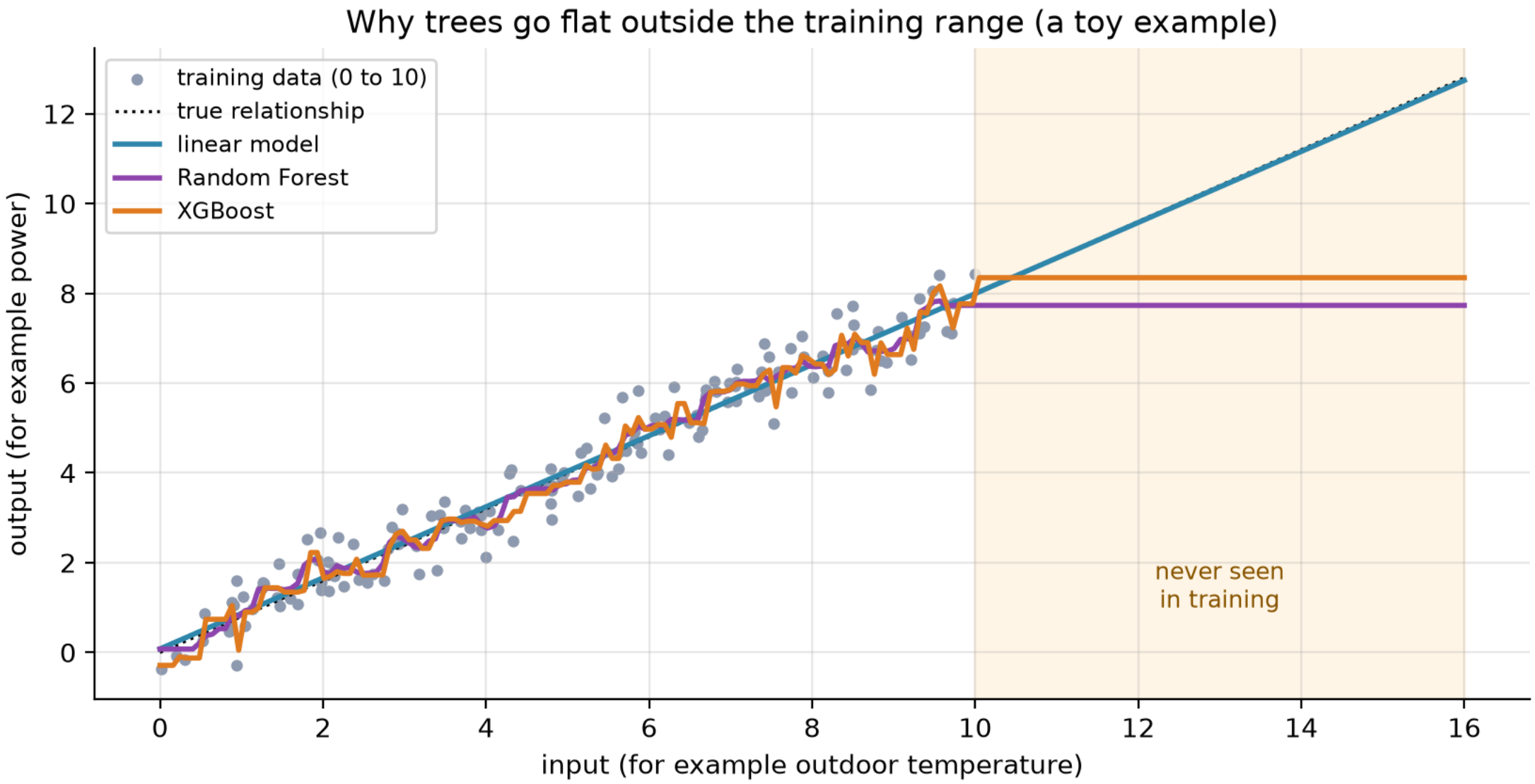
Five kinds of predictor, from a rule of thumb to boosted trees

Stage 6: Model selection
Ankit Kumar

Model	Role in this project	How it fits	Beyond the training range?	Main point
Rule of thumb	No learning	Average by hour; most worn tool first; longest since last purchase	Not applicable	What a company does today
Linear / logistic regression	Statistical baseline	One weight per input, no penalty	Yes (straight line)	Explainable; cannot bend
Ridge	Our model 1	Same, with an L2 penalty chosen by cross-validation	Yes (straight line)	Stable with many inputs
Random Forest	Classic ML baseline	300 trees on random subsets, averaged	No (flat beyond training range)	Few settings; handles thresholds
XGBoost	Our model 2	400 or 300 small trees, each fixing the errors of the previous ones	No (flat beyond training range)	Most accurate on tables

Why trees go flat outside their training range

Stage 6: Model selection
Ankit Kumar



Every model on every test

Stage 6: Model selection
Ankit Kumar

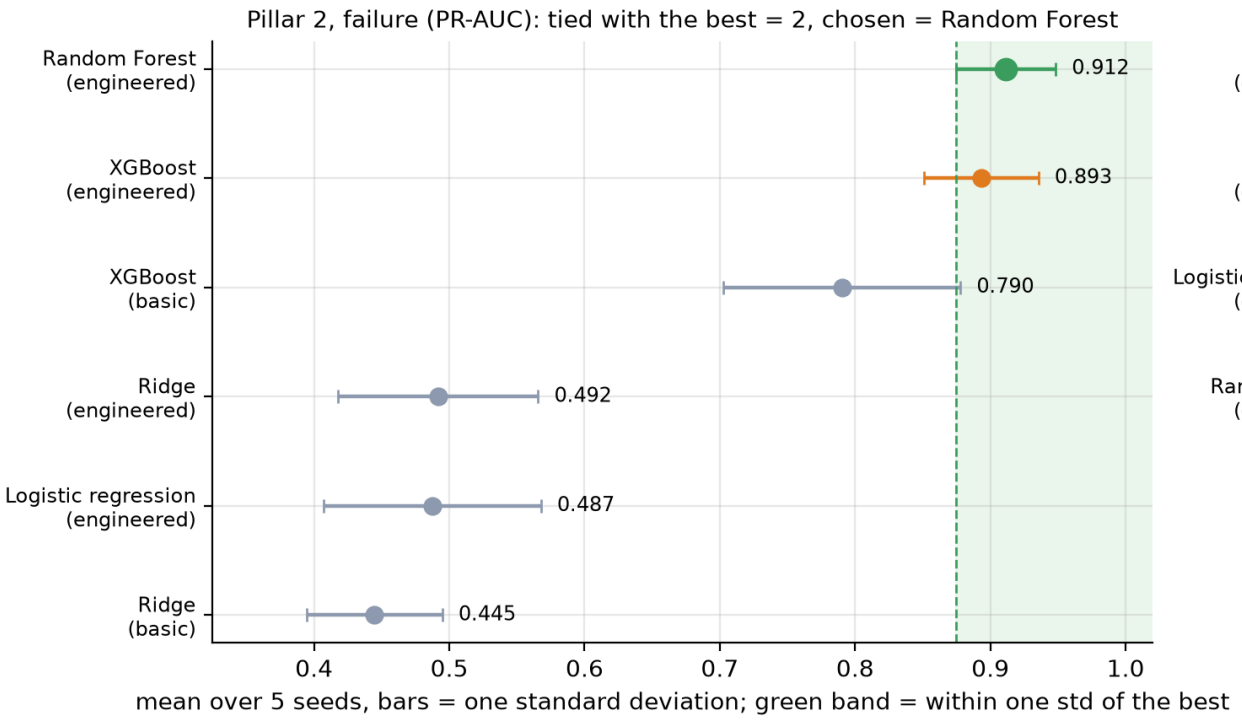
Every model on every test (green = best in its column, red = worst; bold = within 3% of the best)

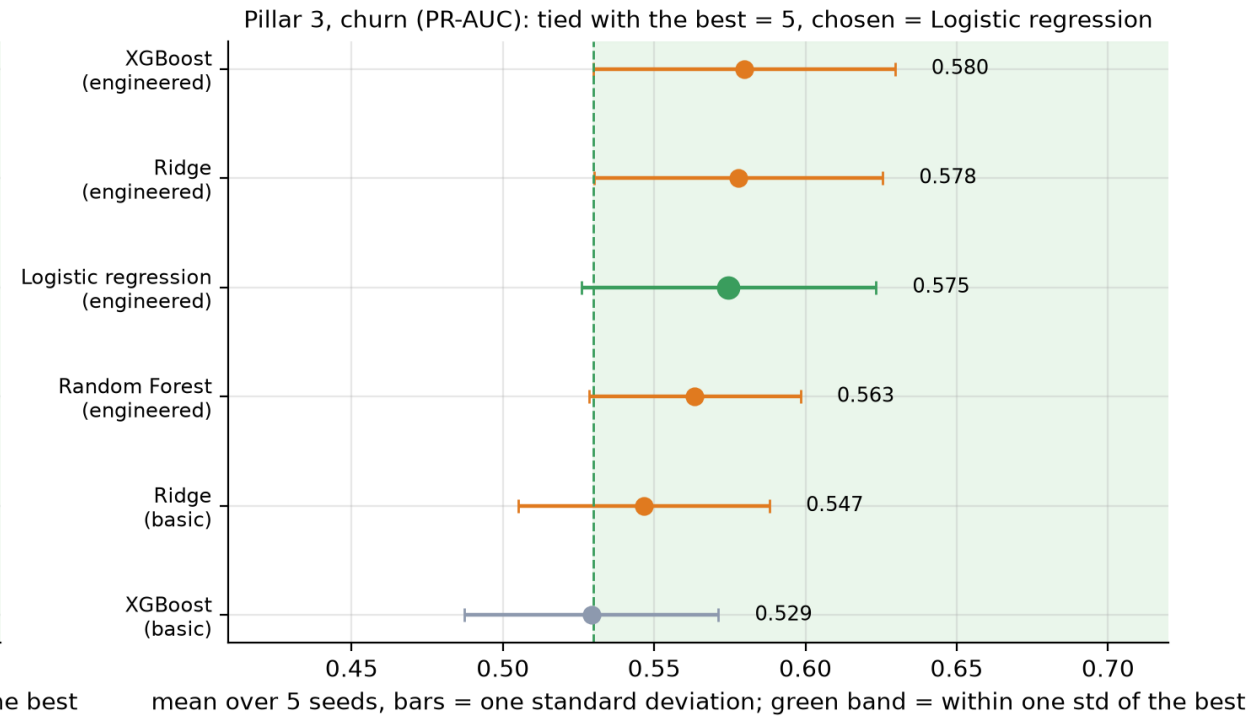
	Demand random	Demand unseen months	Demand forecast	Demand heatwave	Failure PR-AUC	Churn ROC-AUC
Linear / logistic (engineered)	0.096	0.118	0.101	**0.077**	0.49	0.77
Random Forest (engineered)	**0.066**	0.077	0.083	0.365	**0.91**	0.77
Ridge (basic)	0.182	0.224	0.242	0.468	0.44	0.75
Ridge (engineered)	0.096	0.118	0.100	**0.077**	0.49	**0.78**
XGBoost (basic)	**0.065**	**0.070**	0.085	0.332	0.79	0.75
XGBoost (engineered)	**0.065**	**0.072**	**0.076**	0.336	0.89	0.77

Tie rule: the simplest model within one std wins

Stage 6: Model selection
Ankit Kumar

How the tie rule picks a model: the simplest model inside the green band wins



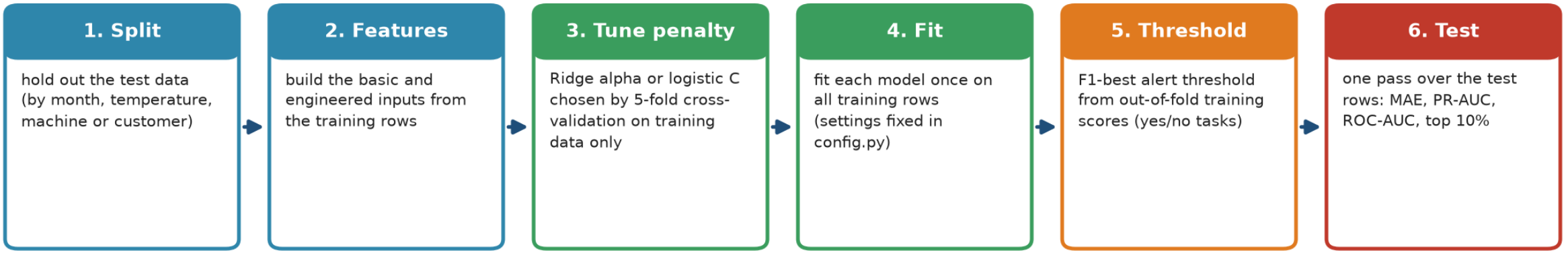


- For failure, Random Forest and XGBoost tie.
- For churn, five models tie, so the simplest, logistic regression, wins.

One protocol for every model, repeated with five seeds

Stage 7: Model training
Animesh Basak

What happens in one run of the experiment (repeated for seeds 42 to 46)



After the six steps the seed is changed and everything is repeated. We report the mean and the standard deviation over the five seeds.

The test rows are used once, at step 6. Nothing is tuned on them.

- Split, features, tune the penalty by cross-validation on training rows, fit once, choose the threshold from out-of-fold scores, test once.

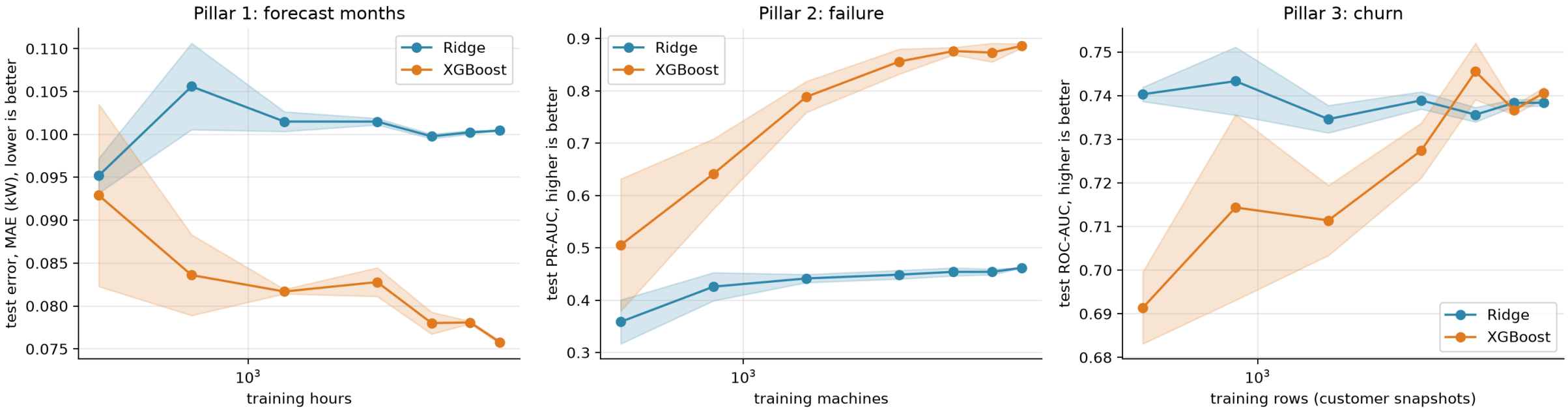
Settings fixed in advance

Setting	Value	Applies to
Seeds	42, 43, 44, 45, 46	All experiments
Held-out share	20% of rows, customers or hours	All pillars
Held-out months / forecast months	3, 6, 9, 12 / 10, 11, 12	Demand
Heatwave threshold	40 °C	Demand
Churn cutoffs	1 Jun, 1 Sep, 1 Dec 2010; 1 Mar, 1 Jun 2011; test 9 Sep 2011	Churn
Ridge penalty candidates	0.001 to 1,000 (alpha); 0.01 to 100 (C); 5-fold cross-validation	Demand; failure and churn
XGBoost, demand	400 trees, depth 4, learning rate 0.05, 80% rows and columns	Pillar 1
XGBoost, failure and churn	300 trees, depth 3, learning rate 0.05, 80% rows and columns	Pillars 2 and 3
Random Forest	300 trees, at least 5 rows per leaf	All pillars
Linear and logistic regression	No penalty	Statistical baseline

Trees need more data than Ridge, and for churn never pass it

Stage 7: Model training
Animesh Basak

Learning curves: test score against the amount of training data (mean of 3 draws, band = one std)

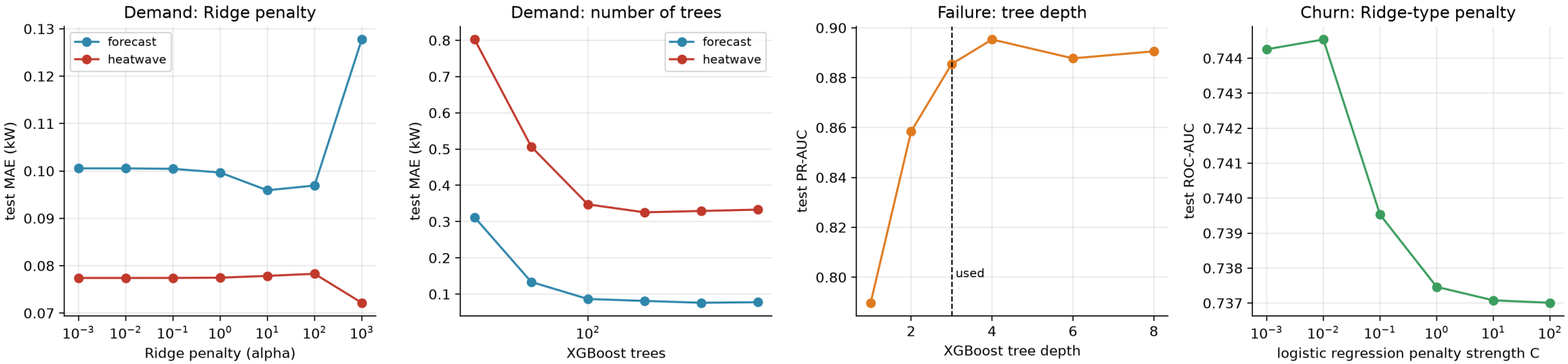


- Ridge learns its straight line early.
- XGBoost climbs for failure and keeps improving slowly.
- For churn both end at about 0.74 ROC-AUC.

The results do not hinge on a lucky setting

Stage 7: Model training
Animesh Basak

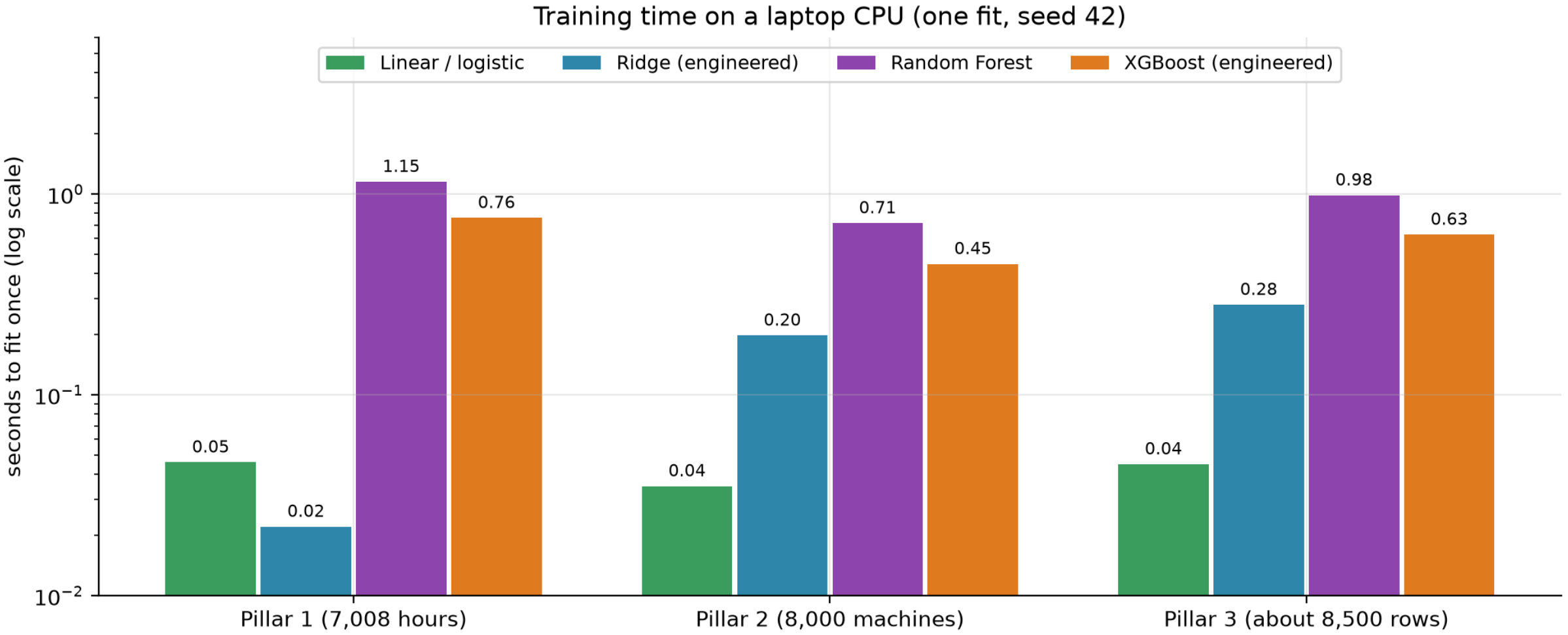
Sensitivity of the main settings: the results do not hinge on a lucky setting



- Ridge penalty, number of trees, tree depth and churn penalty all move the score only a little.
- In the heatwave no number of trees fixes XGBoost.

Training costs seconds, not hours

Stage 7: Model training
Animesh Basak



Demand: error in kW on unseen data

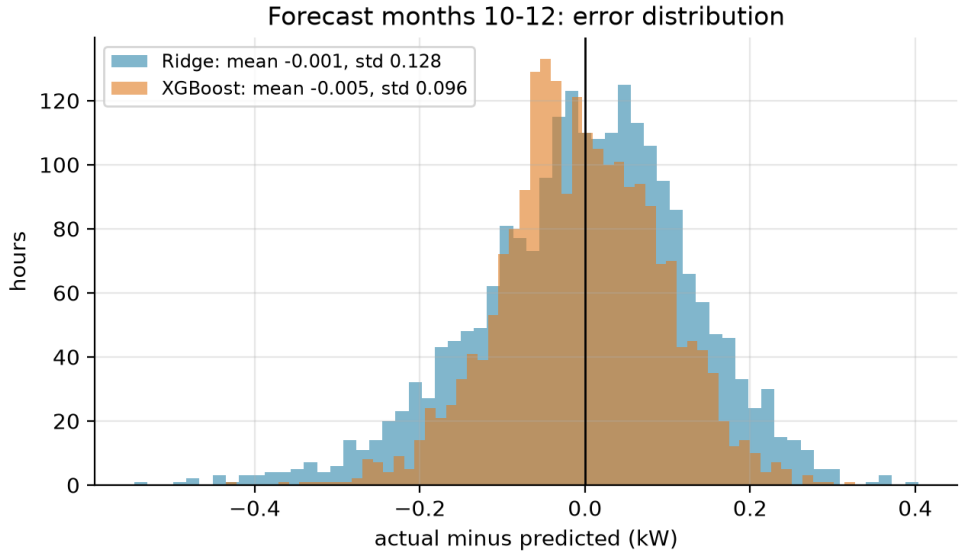
Stage 8: Evaluation and tuning
Animesh Basak

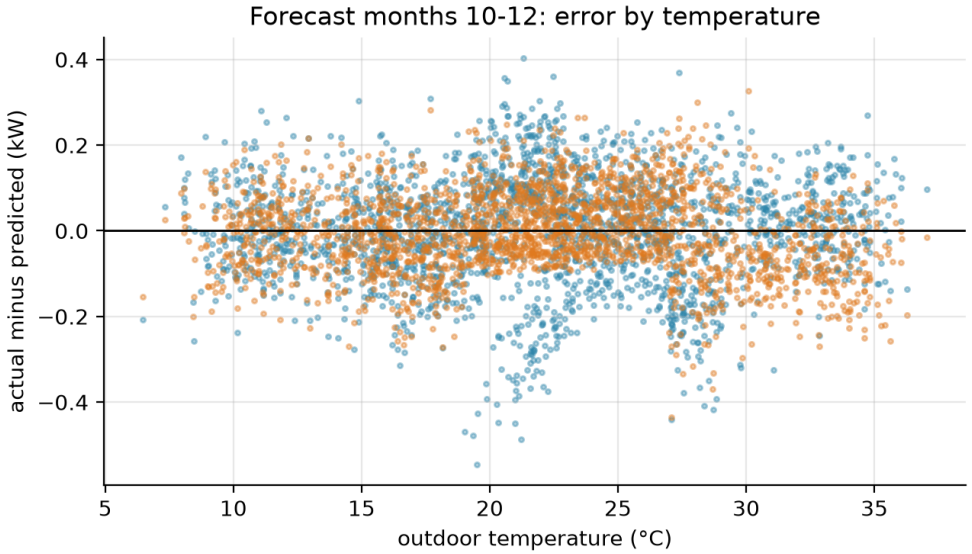
Test	Rule: training mean	Rule: average by hour	Linear regression	Ridge, basic	Ridge	Random Forest	XGBoost, basic	XGBoost
Random 20% of hours	0.942	0.934	0.096	0.182	0.096	0.066	0.065	0.065
Unseen months	1.017	0.995	0.118	0.224	0.118	0.077	0.070	0.072
Forecast, months 10-12	1.012	1.024	0.101	0.242	0.100	0.083	0.085	0.076
Heatwave, 40 °C and above	1.648	1.574	0.077	0.468	0.077	0.365	0.332	0.336

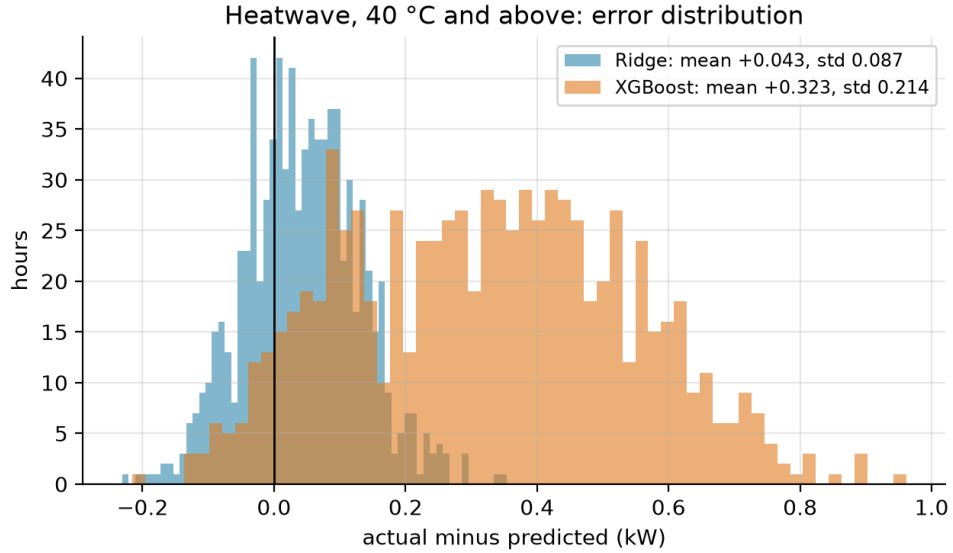
Heatwave: Ridge 0.077 kW, XGBoost 0.336 kW, Random Forest 0.365 kW. Trees cannot predict above the temperatures they saw.

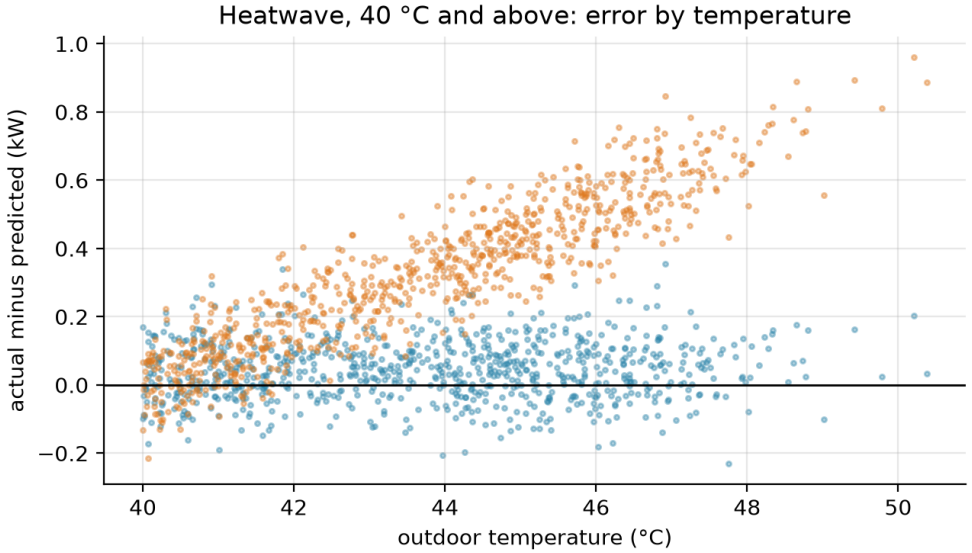
In the heatwave XGBoost under-predicts by up to 0.8 kW

Stage 8: Evaluation and tuning
Animesh Basak



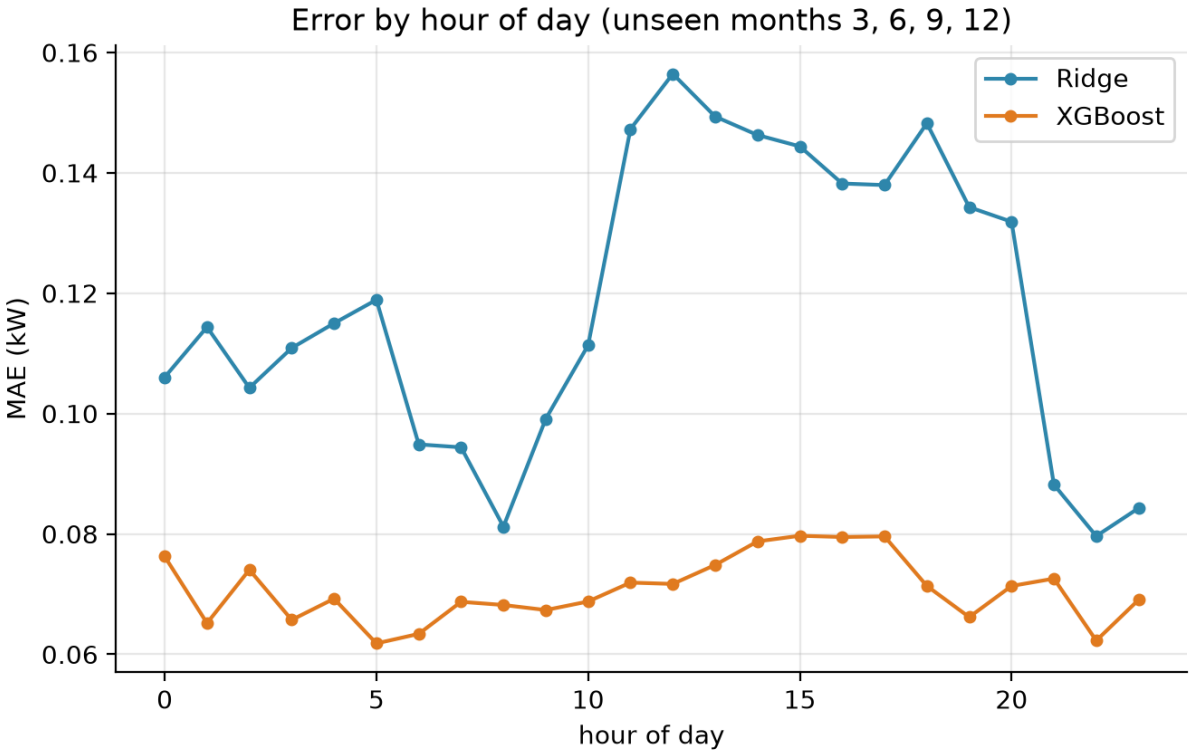






Error by hour and by month

Stage 8: Evaluation and tuning
Animesh Basak





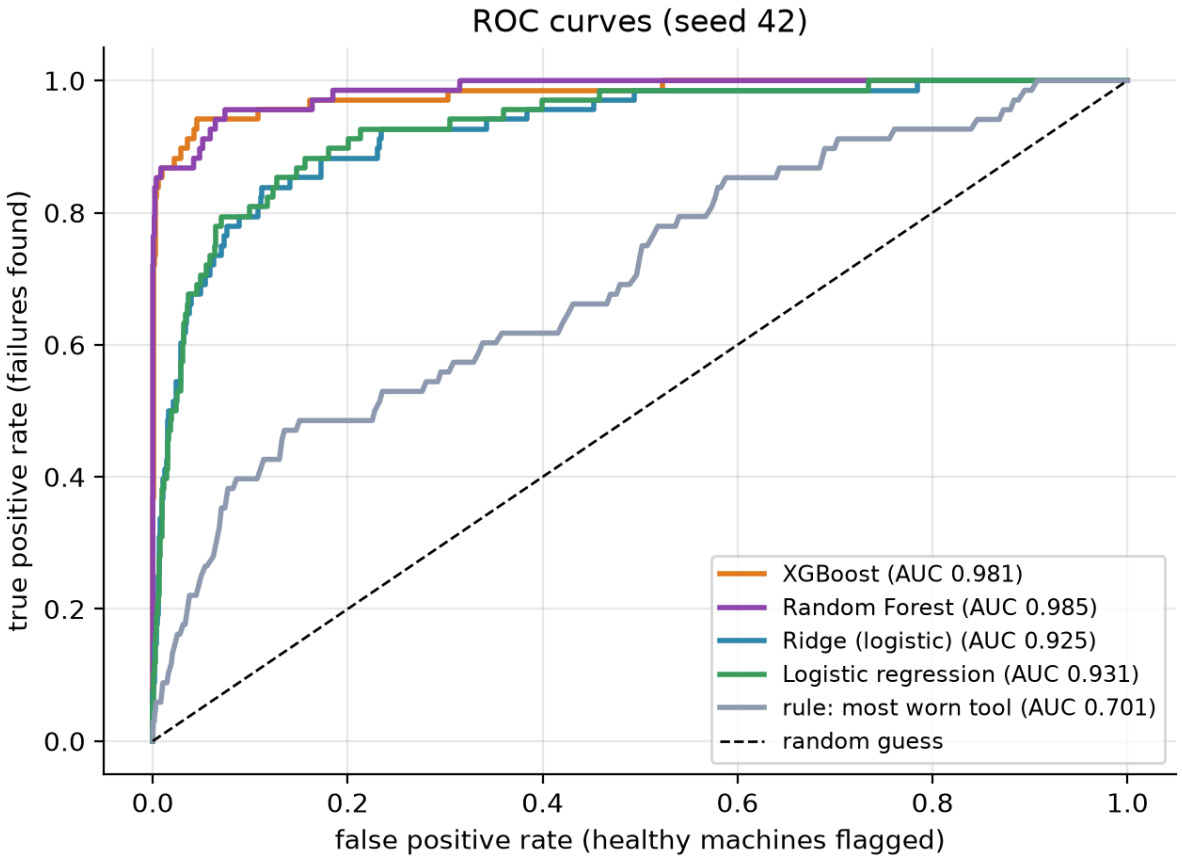
- Ridge struggles in the afternoon and in March.
- XGBoost stays between 0.06 and 0.085 kW in every held-out month.

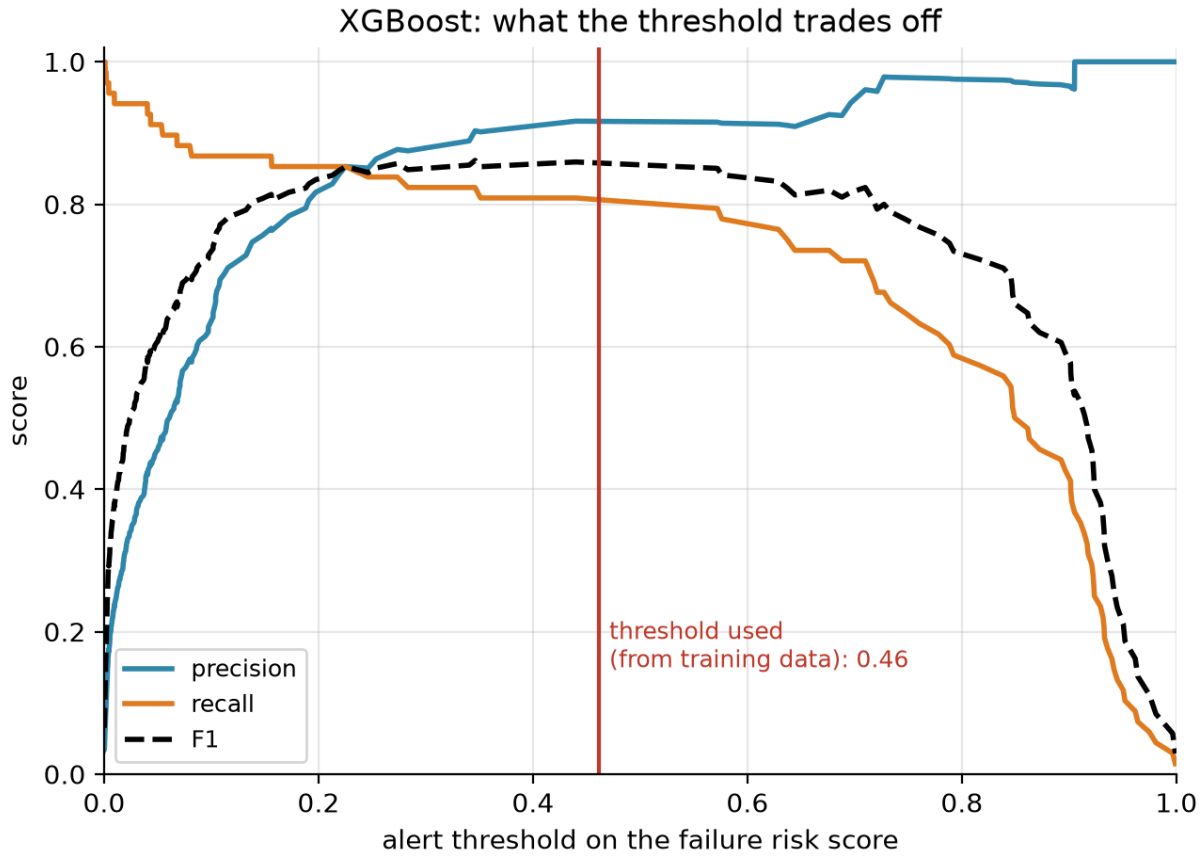
Failure: trees find 94% of failures in the top 10% of machines

Stage 8: Evaluation and tuning
Animesh Basak

Model	ROC-AUC	PR-AUC	F1	Precision	Recall	Failures in top 10%
Rule: most worn tool first	0.655 ± 0.027	0.123 ± 0.013	0.197 ± 0.011	0.158 ± 0.013	0.262 ± 0.012	0.365 ± 0.024
Logistic regression	0.930 ± 0.021	0.487 ± 0.080	0.480 ± 0.059	0.457 ± 0.065	0.512 ± 0.075	0.744 ± 0.069
Ridge, basic	0.898 ± 0.022	0.445 ± 0.050	0.470 ± 0.048	0.456 ± 0.060	0.497 ± 0.085	0.662 ± 0.067
Ridge	0.925 ± 0.021	0.492 ± 0.074	0.489 ± 0.059	0.514 ± 0.088	0.471 ± 0.045	0.732 ± 0.069
Random Forest	0.988 ± 0.007	0.912 ± 0.037	0.887 ± 0.041	0.927 ± 0.046	0.850 ± 0.041	0.941 ± 0.040
XGBoost, basic	0.980 ± 0.009	0.790 ± 0.088	0.727 ± 0.094	0.726 ± 0.138	0.738 ± 0.070	0.912 ± 0.039
XGBoost	0.987 ± 0.007	0.893 ± 0.042	0.851 ± 0.045	0.901 ± 0.039	0.806 ± 0.052	0.938 ± 0.035

Ranking quality and the alert threshold

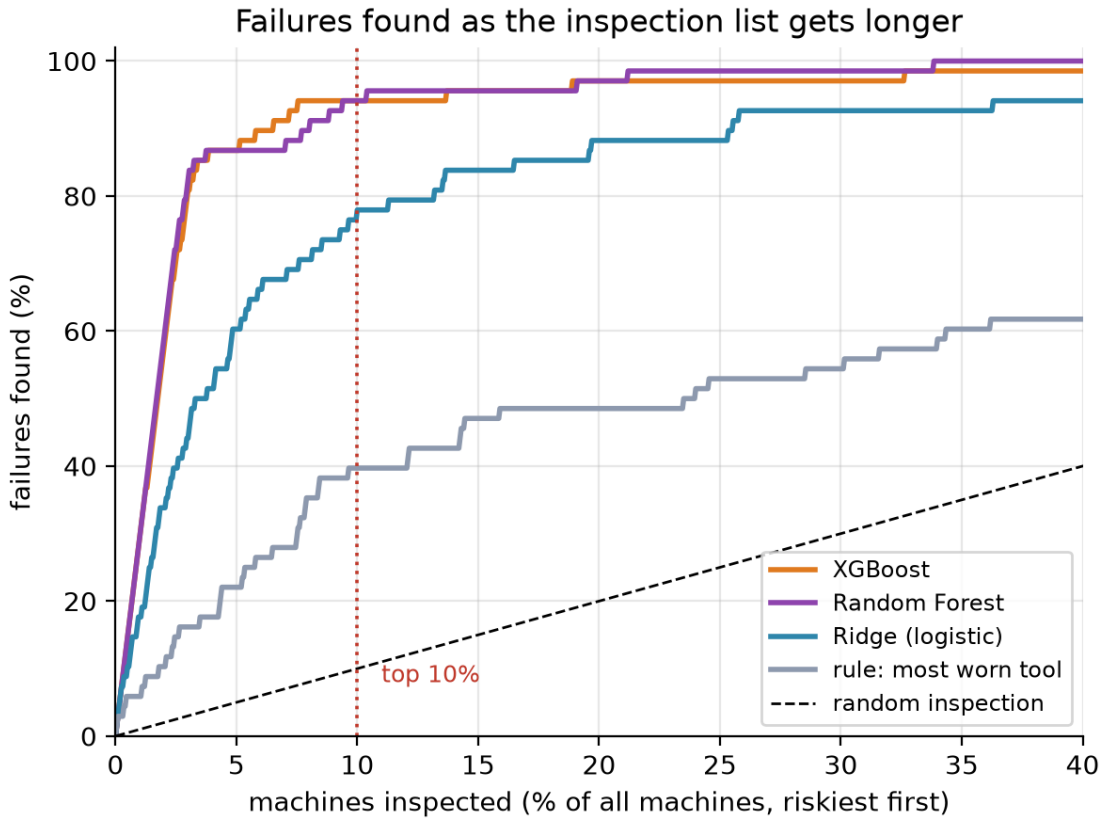


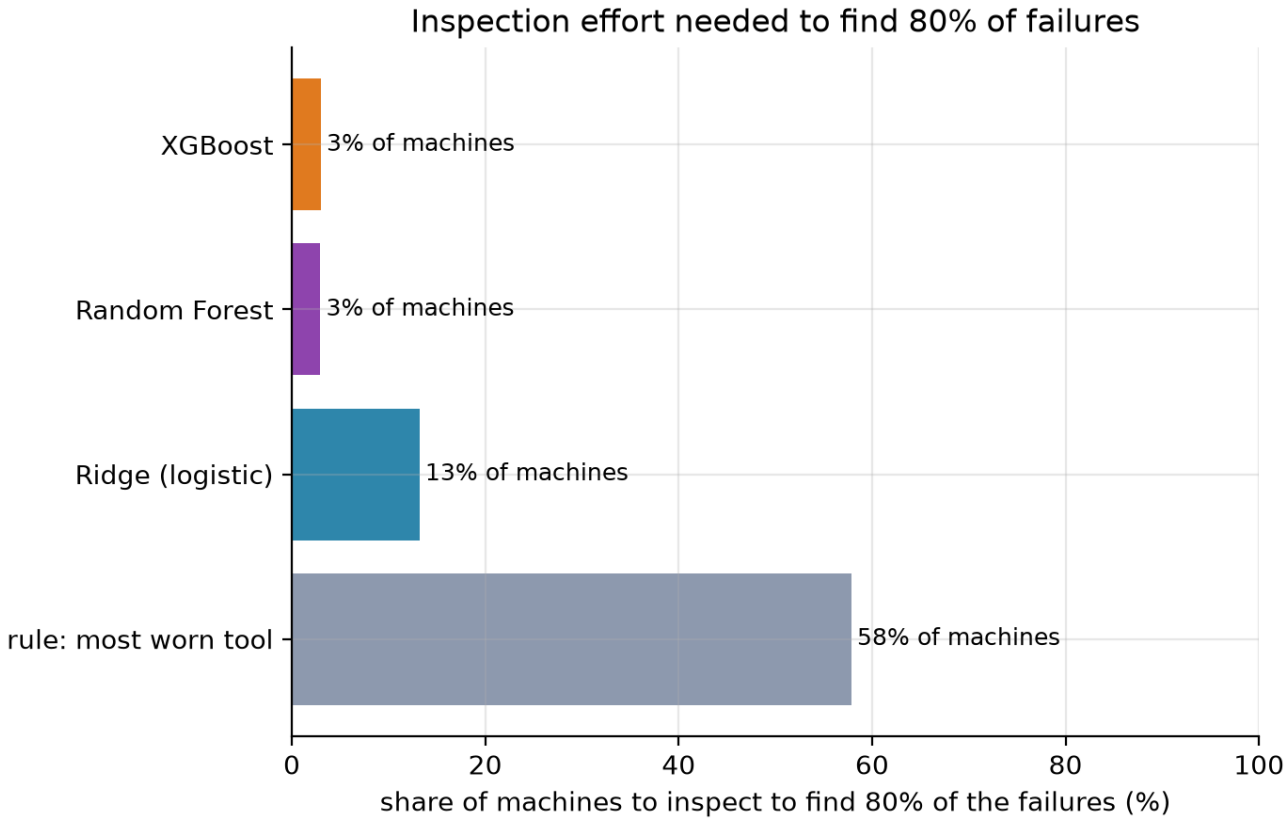


- Random Forest and XGBoost rise almost vertically on the ROC curve.
- F1 is flat from 0.2 to 0.6, so the result does not depend on a precise threshold.

Finding 80% of failures: 3% of machines, not 58%

Stage 8: Evaluation and tuning
Animesh Basak





- The rule of thumb needs about 19 times as many inspections as XGBoost to catch the same share of failures.

Churn: every model reaches about 0.78 ROC-AUC

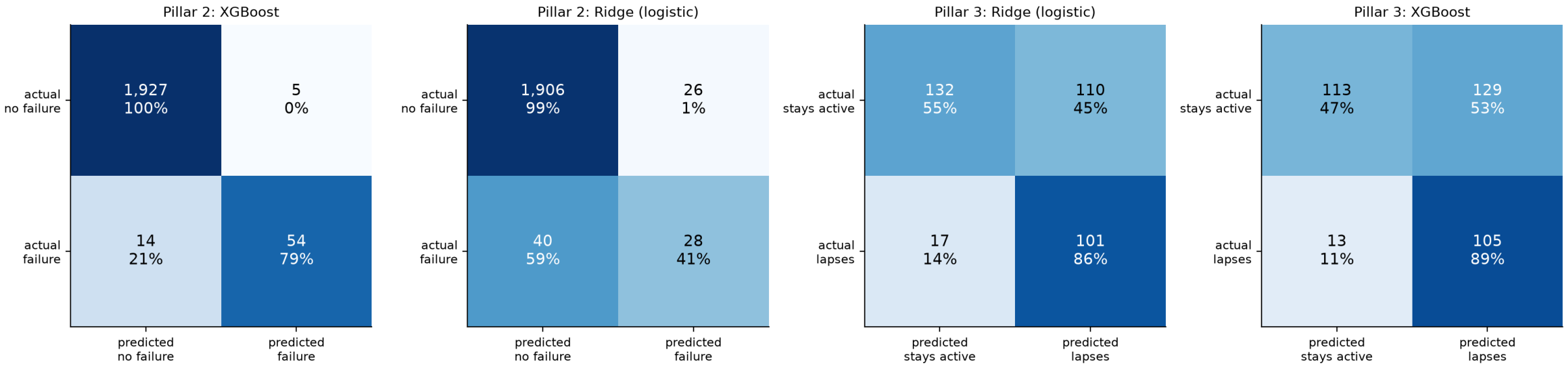
Stage 8: Evaluation and tuning
Animesh Basak

Model	ROC-AUC	PR-AUC	F1	Precision	Recall	Lapsers in top 10%
Rule: longest since purchase	0.607 ± 0.041	0.417 ± 0.019	0.478 ± 0.022	0.318 ± 0.020	0.964 ± 0.019	0.162 ± 0.017
Logistic regression	0.775 ± 0.033	0.575 ± 0.049	0.608 ± 0.024	0.459 ± 0.024	0.903 ± 0.019	0.197 ± 0.021
Random Forest	0.772 ± 0.022	0.563 ± 0.035	0.595 ± 0.017	0.445 ± 0.018	0.901 ± 0.030	0.201 ± 0.025
Ridge, basic	0.750 ± 0.033	0.547 ± 0.041	0.593 ± 0.032	0.476 ± 0.034	0.788 ± 0.039	0.187 ± 0.039
Ridge	0.777 ± 0.032	0.578 ± 0.048	0.609 ± 0.032	0.465 ± 0.035	0.885 ± 0.020	0.202 ± 0.020
XGBoost, basic	0.749 ± 0.031	0.529 ± 0.042	0.590 ± 0.041	0.491 ± 0.039	0.741 ± 0.060	0.178 ± 0.037
XGBoost	0.775 ± 0.029	0.580 ± 0.050	0.609 ± 0.029	0.462 ± 0.032	0.895 ± 0.019	0.200 ± 0.018

Alerts at each model's own threshold

Stage 8: Evaluation and tuning
Animesh Basak

Confusion matrices at each model's own alert threshold (percentages are of each actual row; seed 42, unseen data)

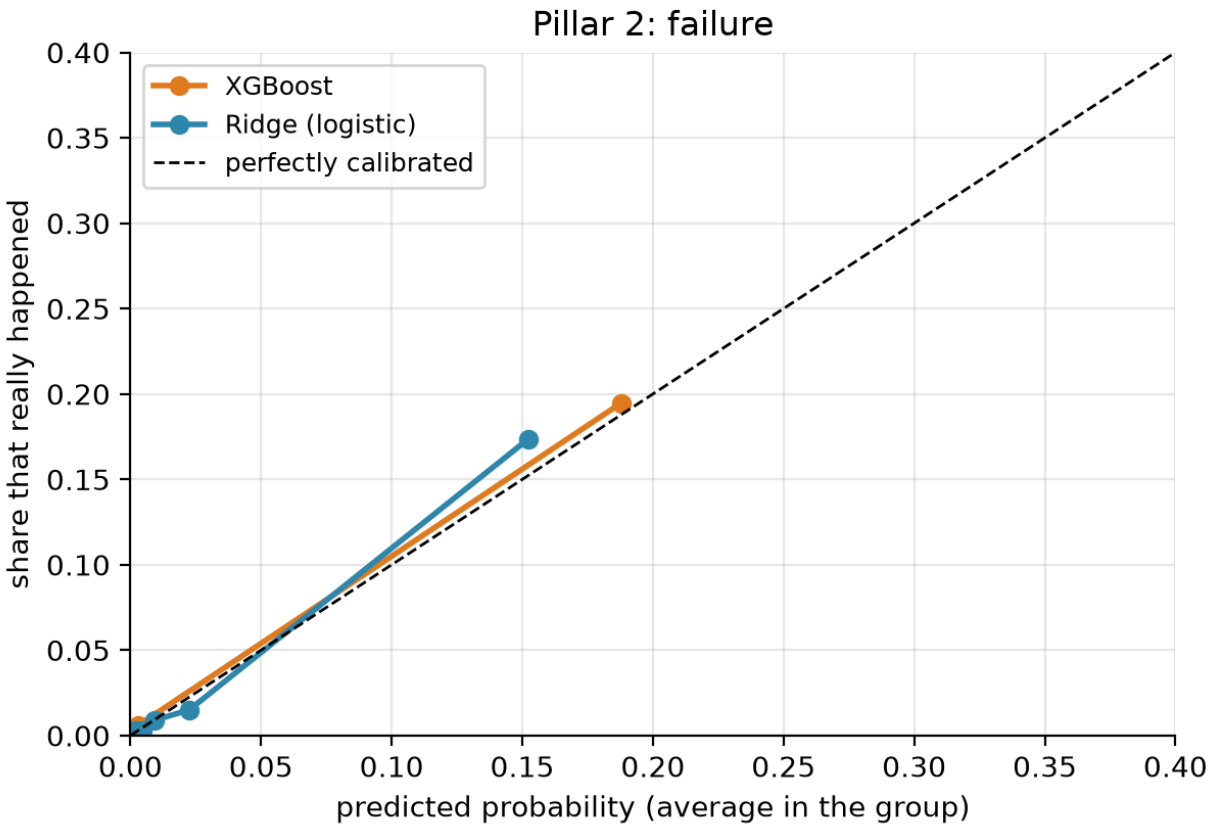


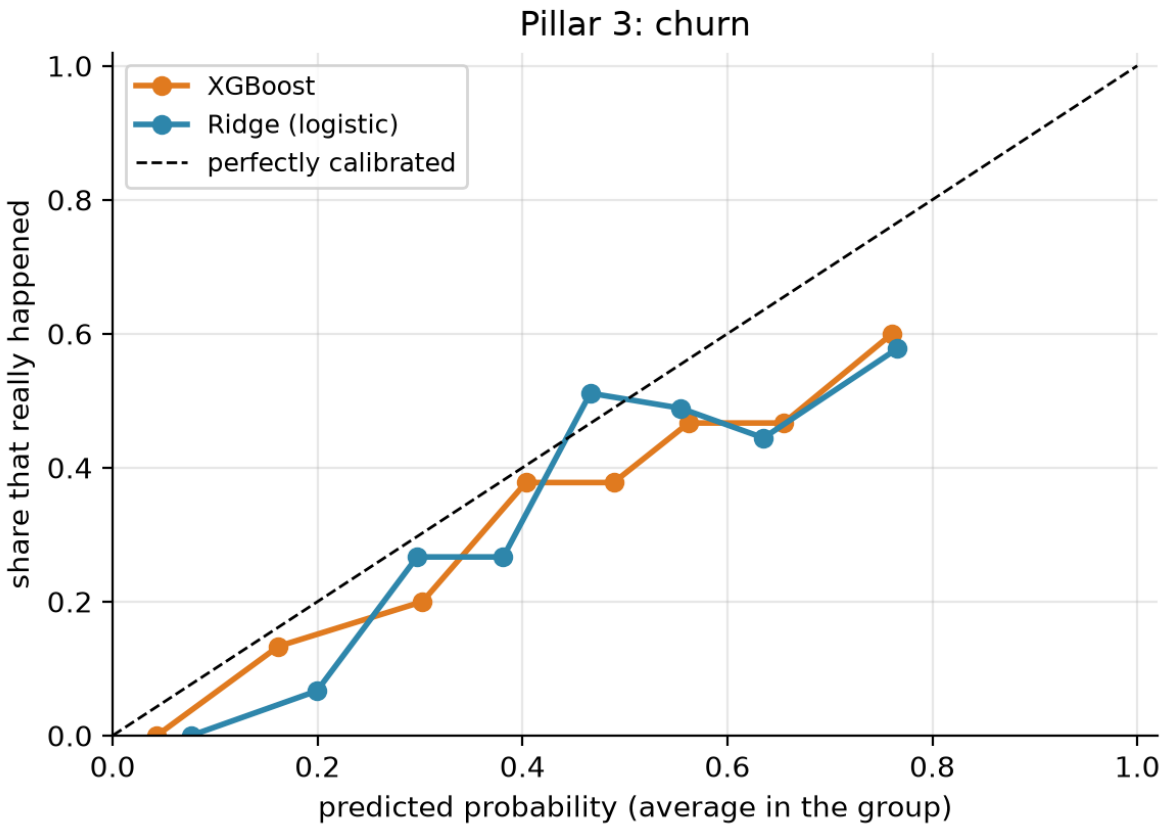
- XGBoost flags 59 of 2,000 machines and 54 are real failures, missing 14 of 68.
- Ridge catches only 28.
- For churn the lists are long, because the F1 threshold favours recall when lapsing is common.

Do the probabilities mean what they say?

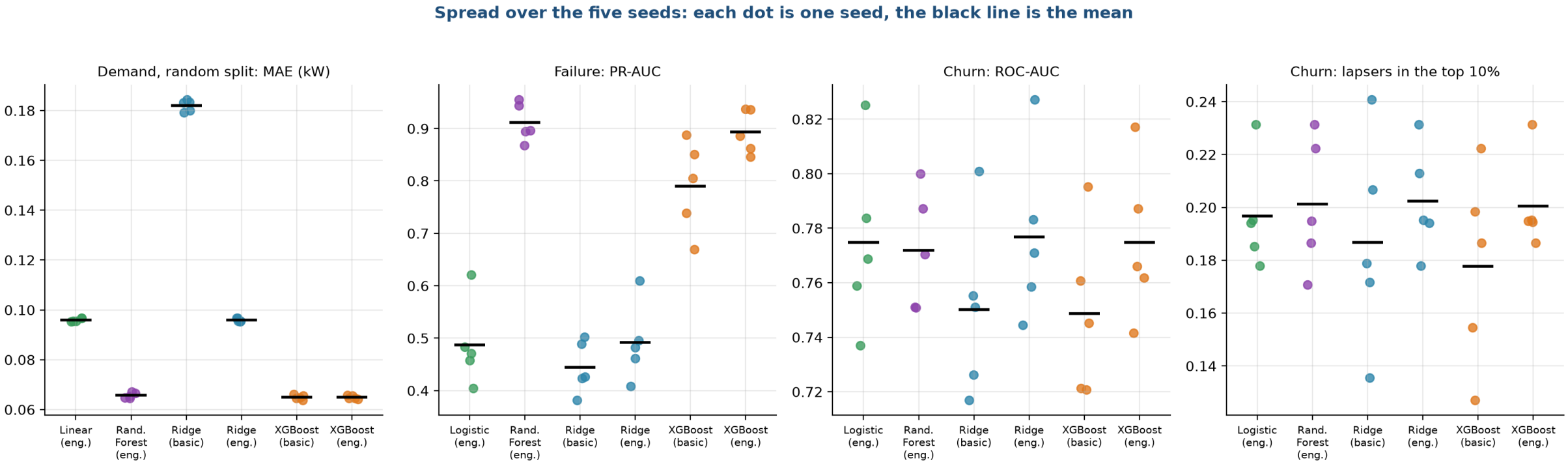
Stage 8: Evaluation and tuning
Animesh Basak

Do the probabilities mean what they say? (groups of equal size, unseen data)





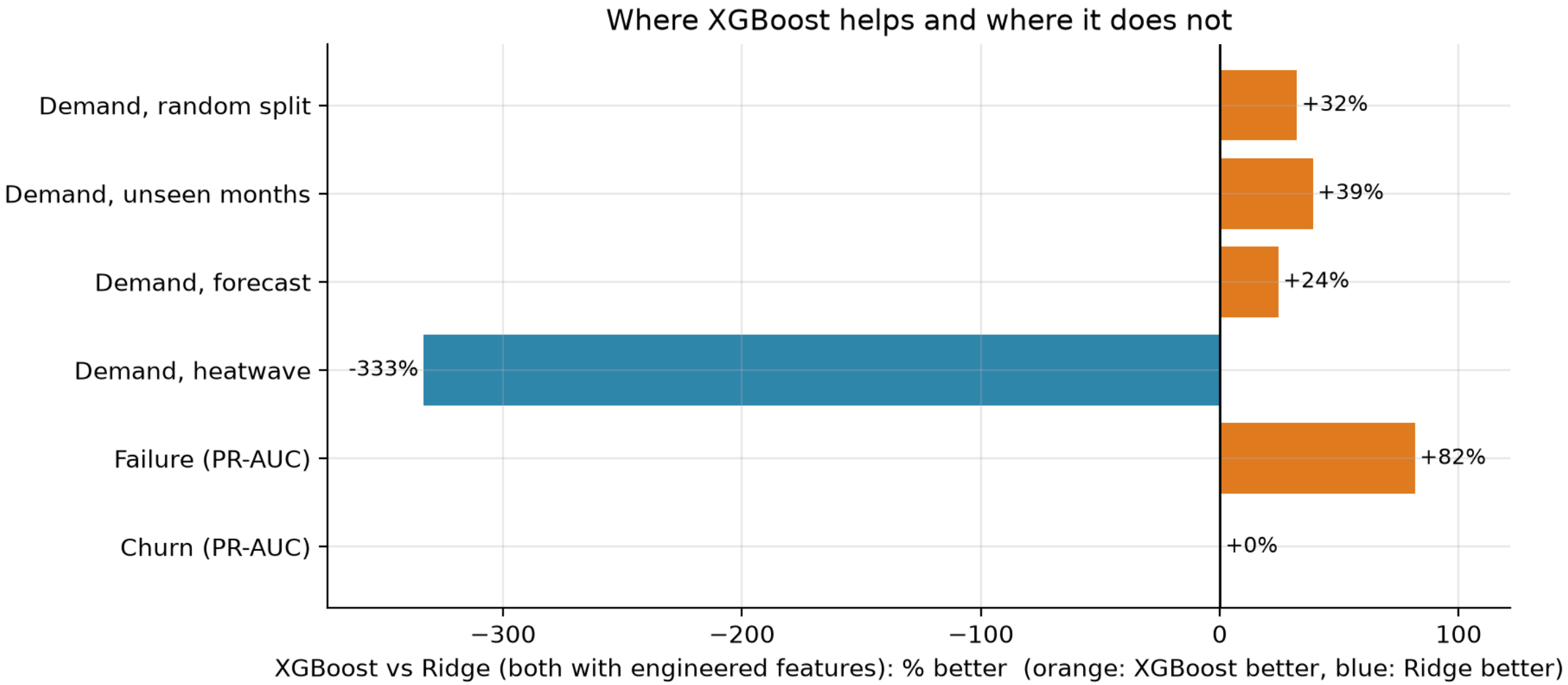
Spread over the five seeds



- Demand barely moves.
- Failure and especially churn vary because the test sets are small, which is why gaps below one standard deviation count as ties.

Where XGBoost helps and where it does not

Stage 8: Evaluation and tuning
Animesh Basak



Scoreboard: the rule of thumb against our models

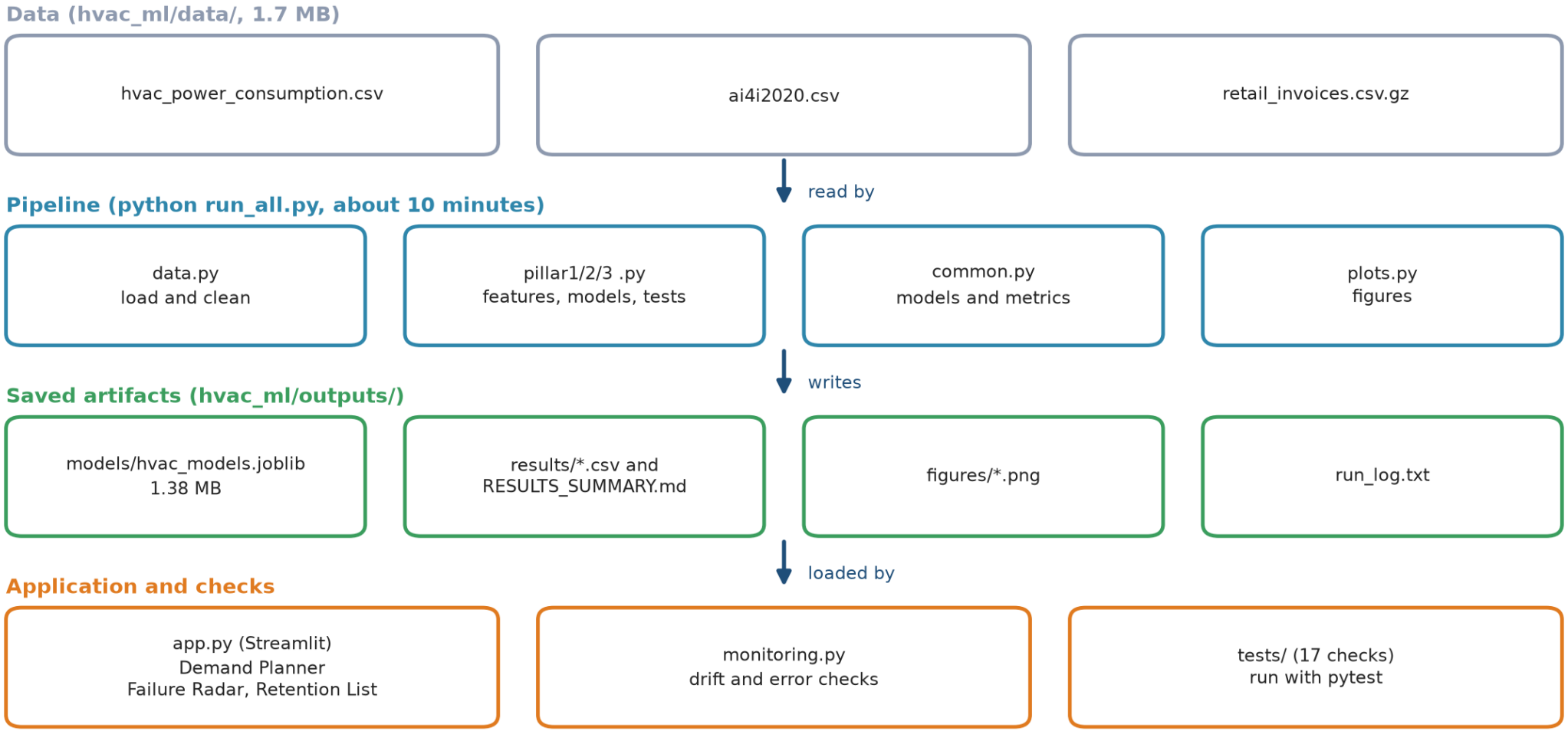
Stage 8: Evaluation and tuning
Animesh Basak

Pillar	Measure	Rule of thumb	Statistical baseline	Ridge	XGBoost	Best result
Demand, forecast test	Error in kW (lower)	1.02	0.101	0.100	0.076	XGBoost
Demand, heatwave	Error in kW (lower)	1.57	0.077	0.077	0.336	Linear models
Failure	PR-AUC (higher)	0.123	0.487	0.492	0.893	Random Forest 0.912, XGBoost close
Failure	Failures in top 10%	36%	74%	73%	94%	Random Forest and XGBoost
Churn	ROC-AUC (higher)	0.607	0.775	0.777	0.775	All four models tie
Churn	Lapsers in top 10%	16%	20%	20%	20%	All four models tie

Architecture: train once, serve in milliseconds

Stage 9: Deployment
Sourav Kumar Boro

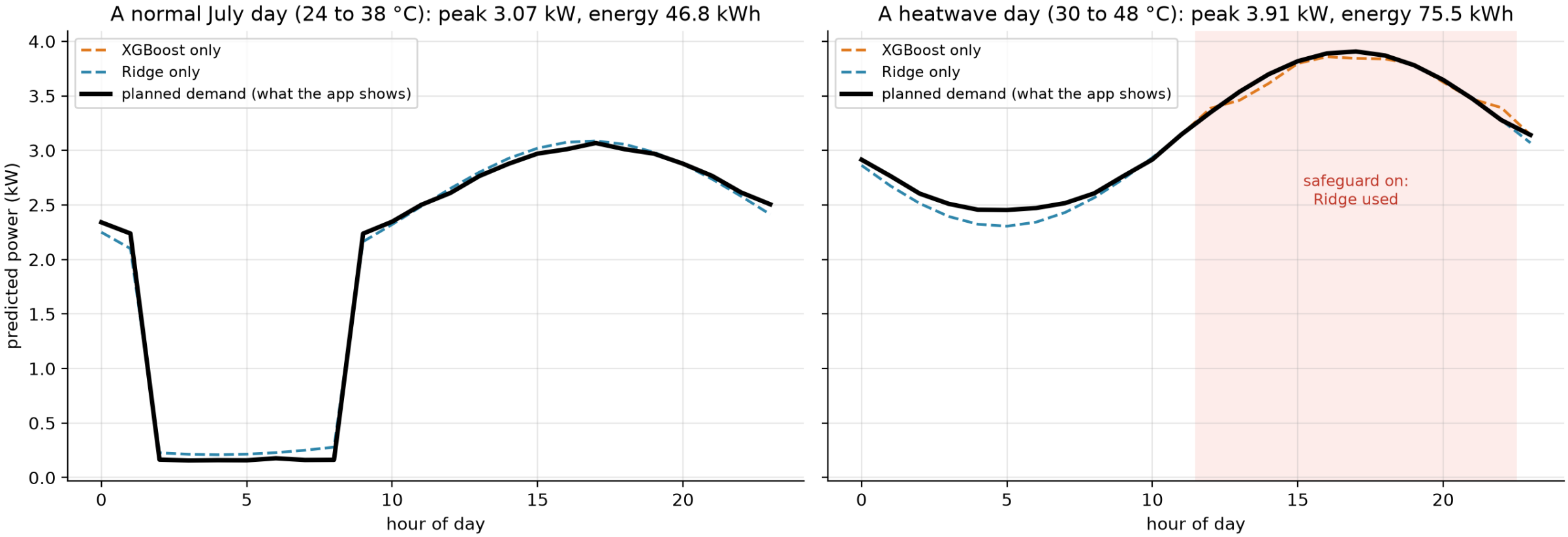
System architecture



users: dispatch manager, maintenance lead, customer-success team

Demand Planner: a normal day and a heatwave day

Stage 9: Deployment
Sourav Kumar Boro

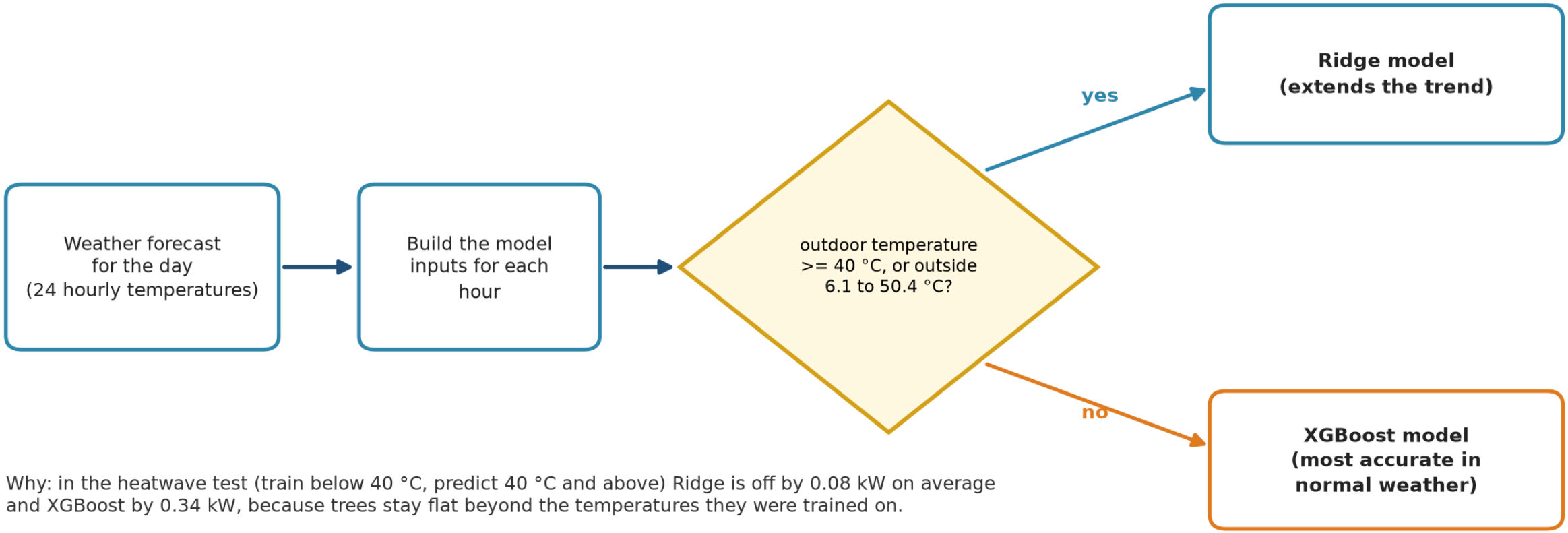


- Peak 3.07 kW and 46.8 kWh on a normal July day.
- Peak 3.91 kW and 75.5 kWh on a heatwave day, where the safeguard uses Ridge from about 11:00 to 22:00.

The heatwave safeguard: Ridge at 40 °C and above

Stage 9: Deployment
Sourav Kumar Boro

Demand Planner: how the heatwave safeguard picks the model for each hour



Why: in the heatwave test (train below 40 °C, predict 40 °C and above) Ridge is off by 0.08 kW on average and XGBoost by 0.34 kW, because trees stay flat beyond the temperatures they were trained on.

The planner shows both curves and says how many hours used the safeguard.

Failure Radar: the 12 riskiest machines in a sample of 500

Stage 9: Deployment
Sourav Kumar Boro

Product ID	Type	Torque (Nm)	Speed (rpm)	Tool wear (min)	Risk (%)	Alert	Actually failed
L51822	L	54.6	1238	226	99.6	inspect	yes
L49778	L	67.5	1320	20	96.4	inspect	yes
M20475	M	71.3	1257	122	96.0	inspect	yes
L53705	L	73.6	1312	49	95.2	inspect	yes
L51800	L	52.6	1336	172	94.6	inspect	yes
M19287	M	48.9	1379	107	94.1	inspect	yes
L51801	L	51.0	1341	174	94.1	inspect	yes
M23874	M	61.3	1340	207	93.6	inspect	yes
M21659	M	12.2	2636	100	93.0	inspect	yes
L51724	L	55.1	1349	187	92.7	inspect	yes
L48763	L	61.0	1310	189	89.7	inspect	yes
L54939	L	59.2	1272	198	89.4	inspect	yes

Illustration: the saved model was trained on all 10,000 machines, so this list is a demonstration, not a test. Honest scores are in stage 8.

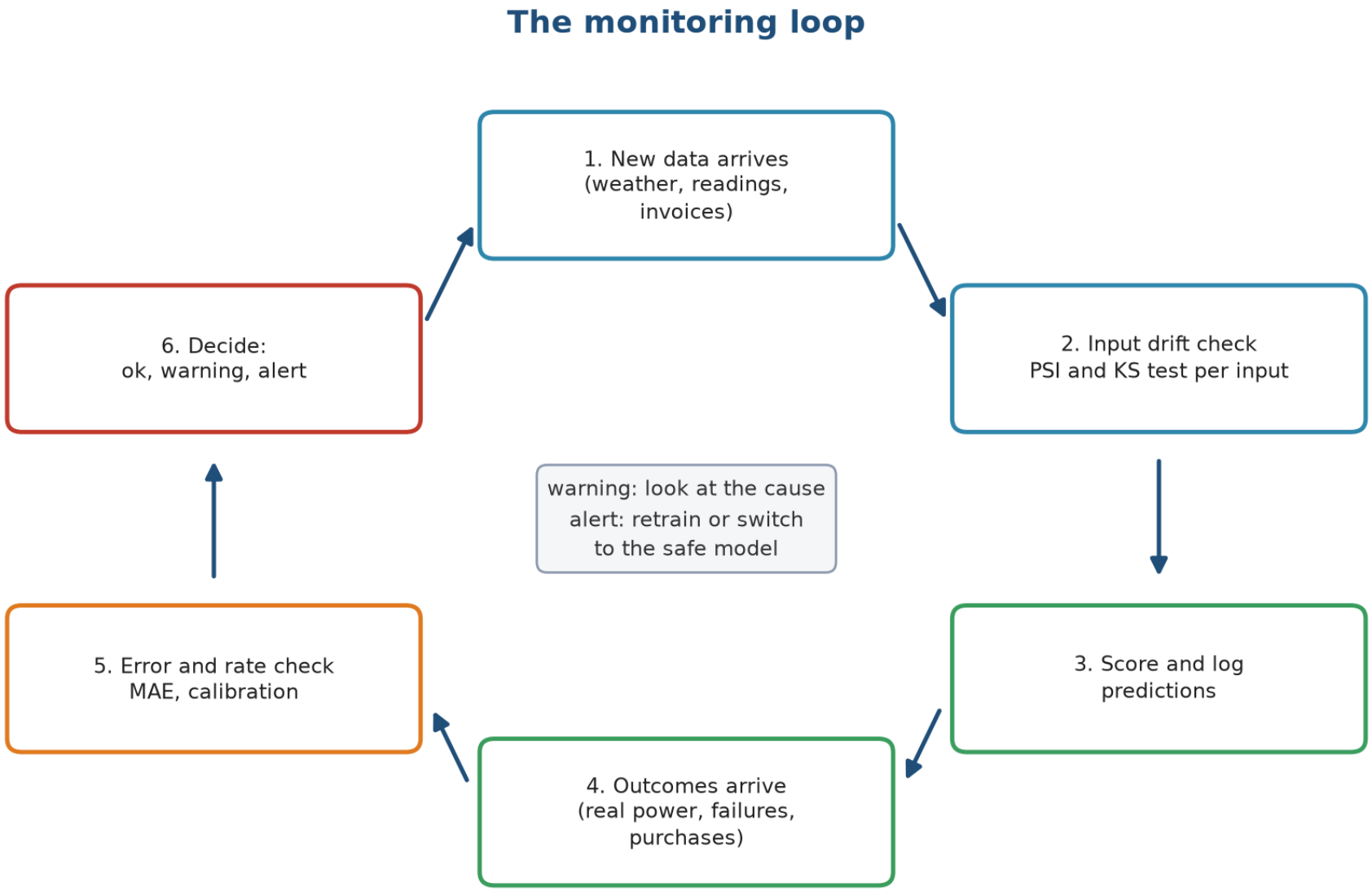
Retention List: the 12 customers most likely to go quiet

Stage 9: Deployment
Sourav Kumar Boro

Customer	Risk (%)	Days idle	Orders	Spend	Orders (90 d)	Actually lapsed
14495	93.0	29	2	92	1	no
17660	91.5	49	2	359	1	yes
14997	90.6	86	2	386	1	no
14171	90.5	6	2	499	1	yes
16641	90.0	13	2	554	1	yes
17958	90.0	24	2	974	1	yes
14079	90.0	32	2	619	1	yes
15374	89.7	36	2	307	1	yes
18281	89.7	88	2	201	1	yes
15256	89.3	56	2	170	1	yes
15219	89.0	84	2	685	1	yes
13302	88.5	63	2	159	1	yes

Illustration: the final model saw these customers. In the honest test, the top 10% of the list reaches 20% of lapsers, twice the random rate.

The monitoring loop



Alert rules

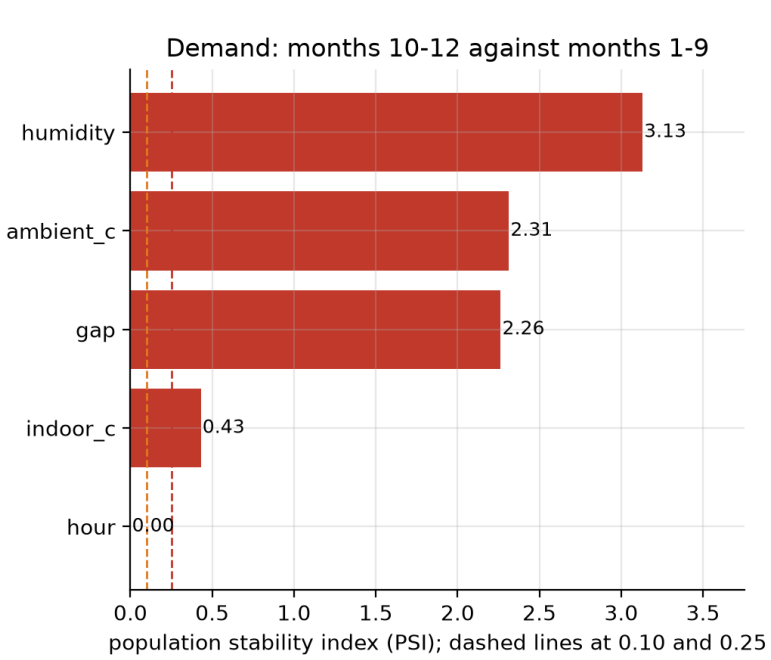
Stage 10: Monitoring
Sourav Kumar Boro

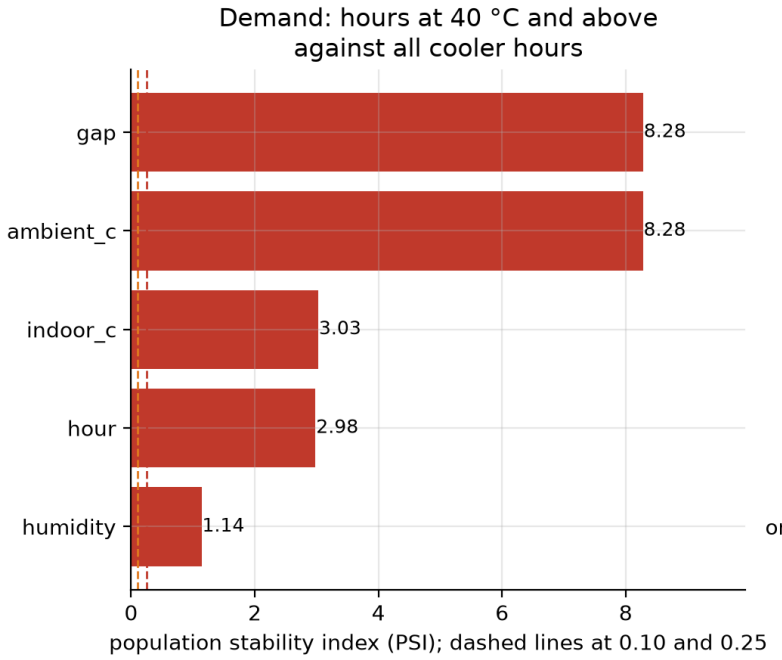
Check	What is measured	Warning at	Alert at	Action
Input drift	PSI of each model input	>= 0.10	>= 0.25	Check the data source; refit if it persists
Demand error	Monthly MAE against the tested 0.065 kW	>= 1.5 times	>= 2.0 times	Refit on the newest months; check sensors
Heatwave safeguard	Hours above 40 °C that used Ridge	any	Error >= 2 x 0.08 kW	Extend the Ridge range; review features
Failure ranking	Failures found in the top 10% (tested 94%)	< 85%	< 70%	Refit with the new failures; review sensors
Churn calibration	Predicted against actual lapse rate	Gap > 0.10	Gap > 0.20	Refit on the latest cutoff; reset the threshold

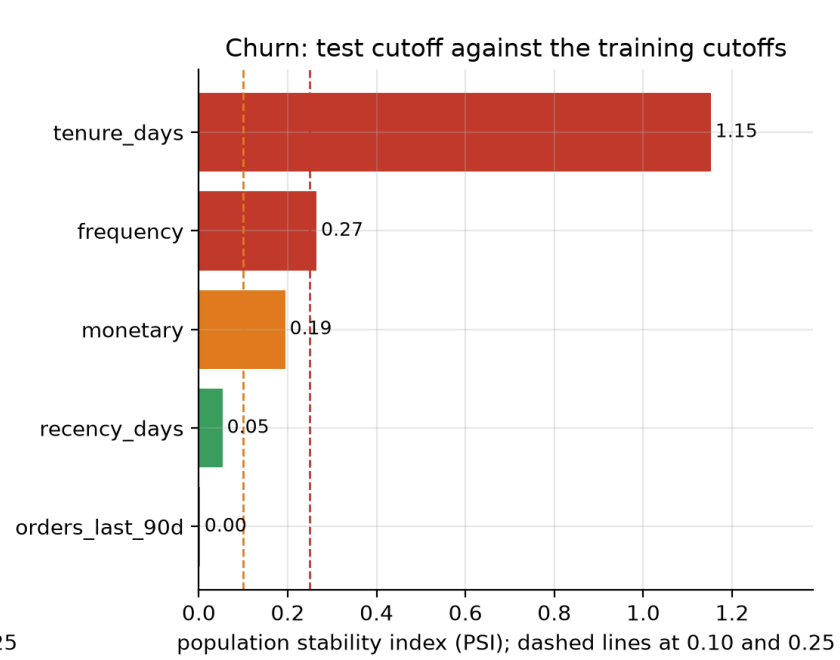
PSI thresholds are the usual rules of thumb (0.10 moderate, 0.25 large shift), not values fitted to our data.

The checks fire when the weather moves into new ground

Stage 10: Monitoring
Sourav Kumar Boro



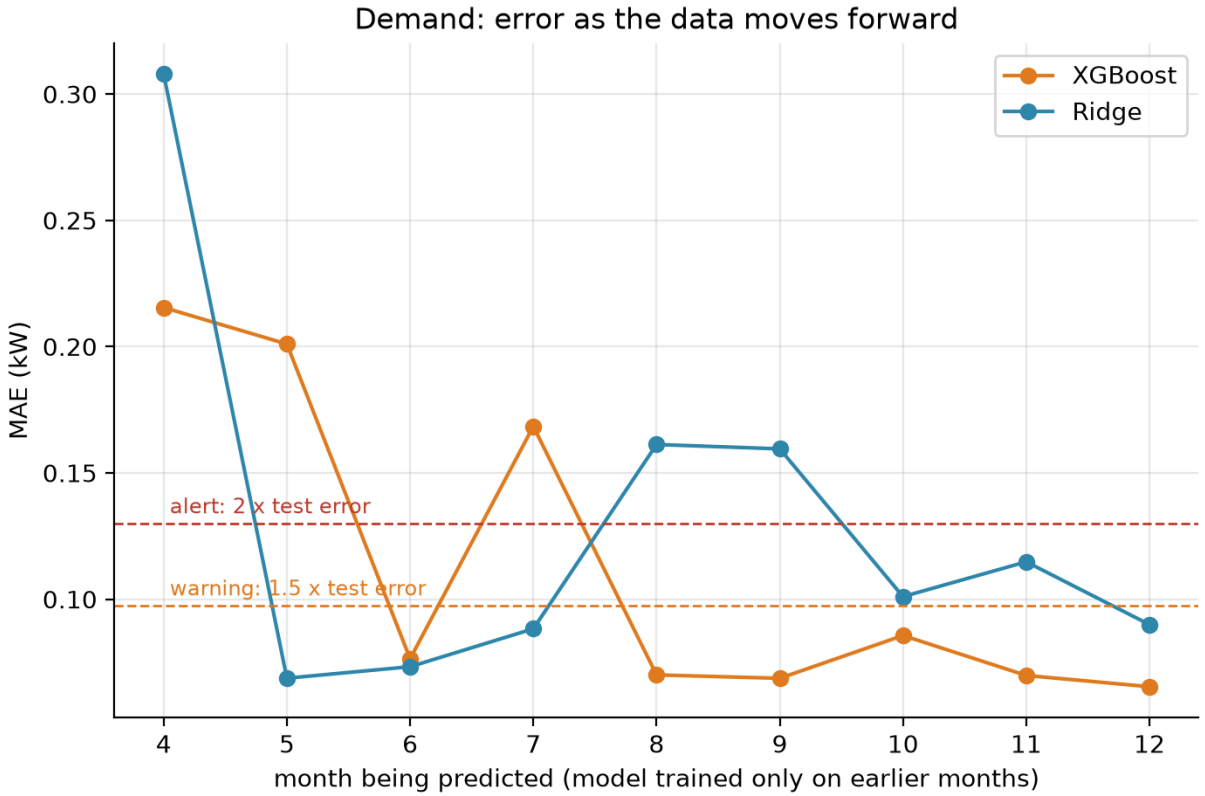


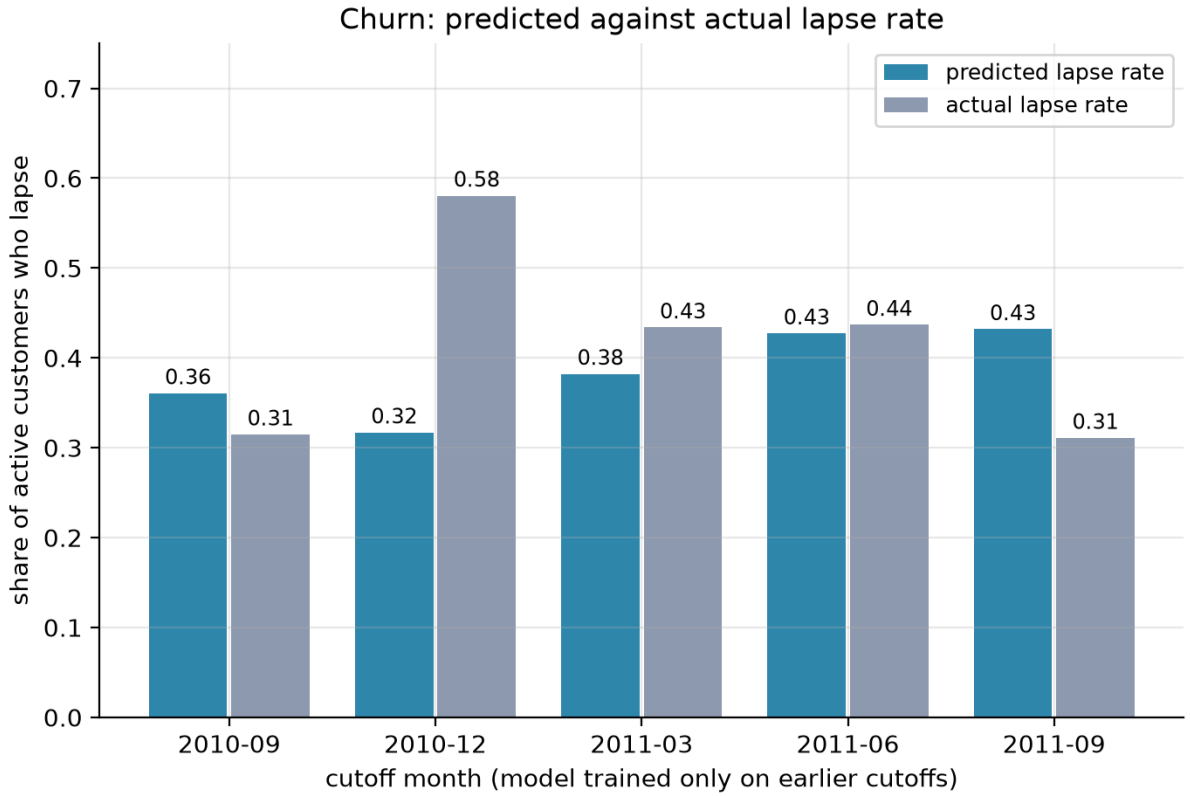


- Months 10 to 12 against months 1 to 9 alert because of the season.
- Heatwave hours against cooler hours reach PSI 8.3.
- For churn, tenure drifts because customers are a year older.

Error over time, and predicted against actual churn

Stage 10: Monitoring
Sourav Kumar Boro

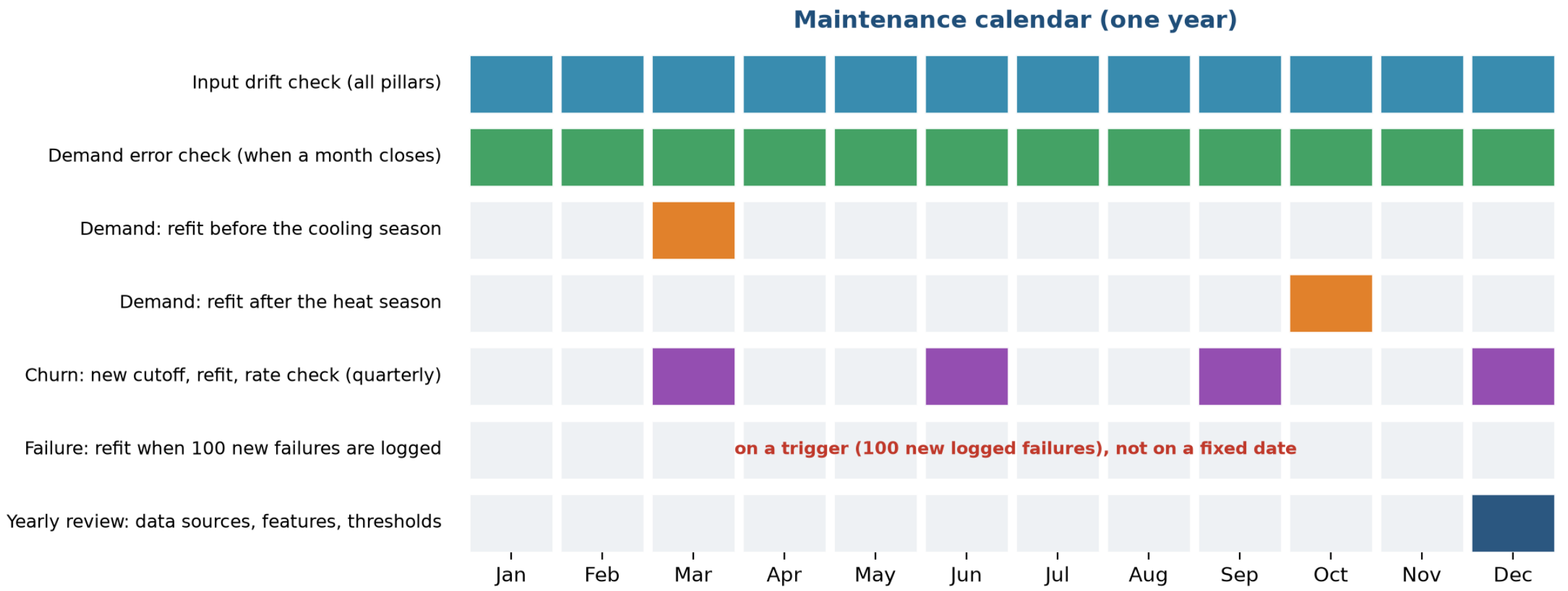




- A model trained only on earlier months has large errors in April, May and July.
- The churn rate prediction misses by 0.26 in the holiday window, although the ranking stays useful.

Maintenance calendar

Stage 10: Monitoring
Sourav Kumar Boro



What each tool is for, and what not to use it for

Summary
Whole group

From result to decision: what each tool is, and what to trust it for

Tool	Model in use and the evidence	Trust it for	Do not use it for
Demand Planner	XGBoost below 40 °C. Ridge at 40 °C and above, or outside 6.1 to 50.4 °C. Error 0.065 kW against 0.93 kW for the rule of thumb. Heatwave error 0.08 kW.	Hourly demand curve, peak hour and energy for the day, to plan staff, stock and grid load.	Planning with XGBoost alone in a heatwave. Other buildings or climates without retraining.
Failure Radar	XGBoost (Random Forest ties). PR-AUC 0.89 against 0.12 for the rule. The top 10% of the list holds 94% of failures, against 36% for the rule.	Ranking machines for inspection and working down the list as far as the team's capacity allows.	Treating the 44% alert level as exact. Trusting the scores on HVAC units before retraining on company data.
Retention List	Ridge (penalised logistic regression). ROC-AUC 0.78 against 0.61 for the rule. The top 10% holds 20% of lapsers: random 10%, rule 16%.	Choosing whom to phone first when the call budget is small.	Forecasting the exact number of customers who will lapse (the rate was off by 0.26 in a holiday window). Expecting more than twice the random hit rate.

- A manager can trust the Demand Planner with its safeguard and the Failure Radar for ranking.
- The Retention List is a modest aid.

Recommendation: pilot Failure Radar and Demand Planner first

Summary
Whole group

Recommendation to an HVAC service provider

Priority	Tool	What to do	Evidence from held-out tests	Condition before relying on it
1	Failure Radar	Adopt. Rank machines by risk and inspect from the top of the list down, within the team's capacity.	Top 10% of the list holds 94% of failures (rule: 36%). Inspecting 3% finds 80% (rule: 58%).	Retrain on the company's own sensor and failure records.
2	Demand Planner	Adopt with the heatwave safeguard. XGBoost in ordinary weather, Ridge at 40 °C and above or outside the training range.	Error 0.065 kW (rule: 0.93 kW). In the heatwave test 0.08 kW with Ridge against 0.34 kW with XGBoost.	Retrain on the company's buildings before each cooling season. Never use XGBoost alone.
3	Retention List	Use as a low-cost aid. Keep the model simple (Ridge).	Top 10% holds 20% of lapsers (random 10%, rule 16%). All models tie at about 0.78 ROC-AUC.	Add product and contract data before paying for a more complex model.
4	Monitoring	Run from the first day. Check input drift monthly, demand error monthly and churn rate quarterly.	Alert at PSI 0.25, error twice the tested 0.065 kW, or a rate gap of 0.20.	Log every prediction and its outcome.

- Retrain on company data before relying on any score.
- Run a four-week side-by-side pilot.

Limits and future work: where a graph model might help

Summary
Whole group

Where a graph neural network might help, and where it would not

Gap in our results	Could a graph model help?	Why it might	Why it might not
Churn: every model ties at about 0.78 ROC-AUC	**Maybe**	Customers who buy the same products may carry signal that one customer's row does not.	The signal may not exist in purchase data. XGBoost with neighbour-averaged features may do as well.
Cross-selling a counter-seasonal appliance	**Maybe**	It is link prediction on a customer-product graph, which graph models learn directly.	Item co-occurrence and matrix factorisation are strong and much simpler.
Demand across several buildings	**Maybe**	Sites on a shared grid or weather system influence each other, and a graph can say so.	We have data for one building only, so there is nothing to test it on.
Heatwave: trees fail above 40 °C	**No**	No reason to expect it.	Trees cannot predict above the range they saw. The fix is a linear model, which the dashboard already uses.
Failure: PR-AUC 0.89 to 0.91	**Not needed**	No gap to close.	The machines in the data are independent rows with no links.

Conclusion

Summary
Whole group

- Machine learning helps most where a ranking drives an action: inspecting machines, with 94% of failures in the top 10% of the list.
- Demand can be forecast to 0.065 kW, but only a linear model stays safe in a heatwave, so the planner switches at 40 °C.
- Retention gains are modest: about twice the hit rate of random calling, and all models tie.
- We recommend a pilot of the Failure Radar and the Demand Planner on company data, with monitoring from the first day.
- All scores come from public stand-in data and unseen test data; the method transfers, the numbers do not.

Thank you. Questions?

References and data sources

Summary
Whole group

Source	Reference
Data 1	felixpradoh. HVAC Power Consumption (hourly, 2025). Kaggle. kaggle.com/datasets/felixpradoh/hvac-power-consumption
Data 2	Matzka, S. (2020). AI4I 2020 Predictive Maintenance Dataset. UCI Machine Learning Repository, 601.
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